Handbook of Finite Fields to 100 IC ideas (2026-11-03)

Handbook of Finite Fields → 100 index-calculus ideas for ECDLP (2026-11-03)

Question. Which mechanisms in the *Handbook of Finite Fields* (Mullen and Panario, eds., CRC 2013; table of contents, chapters 5–17, photographed by the owner) could lower the cost of one phase of an elliptic-curve index calculus, or open a non-generic route, and which of them has this lab already tried?

Status. Catalogue only. Every entry is a *proposal* with a falsifiable claim, a preregistered prediction, the cheapest discriminating test and a falsifier. Nothing here is measured, verified or an improvement; under the success criterion (AGENTS.md) an idea becomes a success only when a frozen, replayed measurement lowers cost against the named IC baseline.

Files

File	What
`IDEAS.md`	the 100 ideas, grouped by IC phase, with handbook section, prior-art status, Lean declaration and gap (generated by `build_report.py`)
`ideas.json`	the same 100 records, machine-readable (`id`, `title`, `ic_phase`, `handbook_sections`, `claim`, `prediction`, `cheapest_test`, `falsifier`, `lean_target`, `prior_art_status`, `merged_from`, `source_page`, `source_idx`)
`pool.json`	the raw mined pool before selection (one record per candidate, with the Haiku prior-art result)
`pages.json`	per photographed page: the sections the miner transcribed and how many ideas it produced
`critic.json`	the completeness critic’s coverage, relabelling, duplicate and weak-claim

File	What
	flags, top ten and missing angles
`formal-results.json`	per Lean batch: author record, verifier record, whether a repair round ran
`formal/`	the core-Lean 4 project `HandbookIC` (toolchain `leanprover/lean4:v4.34.1`, no Mathlib): `Core.lean` vocabulary, `Ideas/B01..B10.lean` claim statements and supporting theorems, `check.sh` and `receipts/`
`diagram.svg`, `phases.svg`	chapter × IC-phase heatmap; ideas per phase with Lean status
`report.html`, `report.pdf`	this README, the two figures and `IDEAS.md`, rendered
`prior-art-index.md`	the lab records the prior-art pass searched
`workflow.js`	the orchestration script (Sonnet miners/selector/formalizers/critic, Haiku prior-art and verifiers)

Method

1. **Mine** (14 Sonnet agents, one per ToC page). Each transcribed the section numbers it could read and proposed 8–10 mechanism-specific ideas tied to a section, with the lab's already-integrated techniques named as off limits.
2. **Prior art** (14 Haiku agents). Each idea was grepped against the lab's IC records listed in `prior-art-index.md` and classified `untried`, `partially-tried`, `tried-open` or `tried-closed` with paths. The cross-repository lead atlas (crypto-autoresearcher) and the handbook knowledge base catalogue (`knowledge/finite-fields/kb`, catalogue not published in this checkout) were not reachable; both remain open checks.
3. **Select** (1 Sonnet agent, structured output validated to exactly 100 distinct ids). Duplicates merged, relabellings and closed leads dropped, phases balanced, ids `H001..H100` grouped by phase.
4. **Formalize** (10 Sonnet agents, 10 ideas each) wrote `formal/HandbookIC/Ideas/Bkk.lean`: for each idea a `def Hnnn_claim : Prop` with every curve-specific fact as an explicit hypothesis, plus at least one proved supporting theorem. **Verify** (Haiku per batch): compile, forbidden tokens, `#print axioms`, claim declarations present; one Sonnet repair round for failures, then re-verification.
5. **Critic** (1 Sonnet agent): coverage, relabelling, duplicates, weak falsifiers, top ten, missing angles.
6. Native evidence: `formal/check.sh` (full `lake build`, token scan, axiom audit, `receipts/`) and the CI gate `tools/lean/current-corpus-gate.py`, extended in this change to admit registered dependency-free core-Lean projects (`CORE_LEAN_PROJECTS`), with refusal controls in `tools/lean/test_current_corpus_gate.py`.

Formalization scope (partial, by design)

The Lean corpus models the exponent side only: a `DLPInstance` is a type with a log bijection onto `Fin n`; factor bases, relations, smoothness counts, orbit quotients and phase costs are abstract. The elliptic-curve group law, field arithmetic, summation polynomials and character sums are **not** modelled, so each `Hnnn_claim` carries those facts as hypotheses and each record names its `missing_bridge`. The theorems are cost-model and counting lemmas (no `sorry`, standard foundations only). Mathlib was not used: the session's disk allowance could not hold the pinned Mathlib cache, and the lab's Mathlib-pinned Hecke workspace is the route for any idea that advances to a measured result. This is the "partial formalization" case of the Lean rule in AGENTS.md: the full elliptic-curve statements stay open.

Results

- **Pool**: 113 raw ideas from 14/14 pages (8, 8, 8, 8, 8, 8, 8, 8, 8, 9, 8, 8, 8, 8 per page). Prior-art pass over the pool: partially-tried 85, tried-closed 11, tried-open 5, untried 12.
- **Selected**: 100 ideas, ids H001–H100. Per phase: factor-base 18, decomposition 25, relation-collection 12, relation-filtering 8, linear-algebra 4, descent 7, cost-model 9, curve-selection 6, hardware 11. Prior-art status: partially-tried 80, tried-closed 4, tried-open 4, untried 12. Selector notes on what was dropped: `dropped-summary.txt`.
- **Handbook chapters cited** (an idea may cite several): ch 5: 10, ch 6: 11, ch 7: 6, ch 9: 15, ch 10: 16, ch 11: 18, ch 12: 19, ch 13: 17, ch 14: 26, ch 15: 19, ch 16: 24, ch 17: 10. Heatmap: `diagram.svg`.
- **Lean**: 10/10 batch modules compiled and passed the Haiku verifier on the first round (no repair round ran); 100/100 ideas have at least one proved supporting theorem (426 theorems named by the authors). `formal/check.sh`: full `lake build` clean, no forbidden token, **462 theorems audited**, none outside `propext`, `Classical.choice`, `Quot.sound` (`formal/receipts/`). The CI gate's new checker was run natively on every tracked module and admitted all of them.

What the critic found (adversarial pass, kept verbatim in `critic.json`)

The catalogue has 100 ideas, and three things limit it. First, coverage is lopsided: decomposition (25) and factor-base (18) hold 43 of the 100, linear-algebra has 4 (2 of them Spot logistics), and descent has 7. Second, many ideas measure a distribution or predictor rather than a changed algorithm: about 12 are best-of-T subspace or coset selection (H001,H004,H008,H009,H010,H013,H014,H015,H017,H042,H072,H073). A13 already measured coset selection as near-null, and their own predictions shrink to about 1.0x at production size. Third, about 8 are closed or admitted-null re-asks (H002,H020,H078,H083,H088,H089,H100,H065). I flagged 18 relabels, 27 duplicates (several of them chains, such as H015, H044, H024/H026 and H097/H099), and 36 weak ideas. Relabel examples are H050, H090, H086, H084, H076, H062, H064/H093, H095, H077. The strongest first-improvement candidates change something countable and have a checkable output: group-structure quotients (H044, H007), early stopping (H045), identical-output hot-loop rewrites (H026, H097, H098), the cofactor factor base (H012), core extraction before Wiedemann (H059), a 33% matvec cut (H066), and isotropic F4 pairs (H041). Merge the duplicate chains before spending runs. Add LA, prime-field, large-prime accounting and scale-extrapolation ideas. Every winner must report the gap to the matched rho with the section 5a pricing band.

- Relabelling flags (18): H050, H090, H086, H084, H076, H062, H074, H095, H064, H093, H077, H058, H083, H075, H078, H009, H063, H048. These re-account or rename a cost rather than change the algorithm; under the success criterion they cannot be improvements and stay only as accounting checks.
- Duplicate flags (27): H092 → H063, H048 → H039, H014 → H004, H042 → H013, H008 → H004, H010 → H015, H009 → H015, H001 → H015, H073 → H015, H072 → H013, H017 → H015, H054 → H044, H087 → H044, H031 → H023, H026 → H024, H049 → H045, H059 → H057, H058 → H057, H053 → H046, H099 → H097, H043 → H025, H100 → H018, H082 → H080, H041 → H038, H029 → H075, H066 → H065, H002 → H088. A later pass should merge these chains before any is scheduled.
- Weak-falsifier flags (37): H020, H100, H065, H009, H010, H015, H017, H073, H072, H001, H002, H078, H083, H088, H089, H077, H080, H082, H079, H064, H093, H095, H091, H094, H096, H081, H068, H069, H070, H011, H021, H067, H004, H014, H051, H085, H074.
- Missing angles named by the critic:
  - Linear-algebra is thin (4 ideas, 2 of them Spot logistics H064/H093). There is nothing on Pohlig-Hellman/CRT interplay with small prime factors of the group order, lower-precision modular arithmetic, structured elimination before Lanczos, or LA for the m=2 low-weight rows.
  - Decomposition and factor-base hold 43 of 100 ideas, and roughly 12 of them are variants of best-of-T subspace or coset selection (H001,H004,H008,H009,H010,H013,H014,H015,H017,H042,H072,H073). A13 already measured coset selection as near-null. This is oversupplied and unlikely to produce a first improvement.
  - Prime-field curves (ECCp-131, Curve25519-class) are covered only by H017, H044, H054, H087 and H097. The Weil-restriction, F4, and subspace ideas are binary-specific, and no idea covers prime-field decomposition (for example Gaudry/Diem-style degree-2 and index-calculus-for-prime-field-extension).
  - The large-prime / double-large-prime relation cost model has no idea, although large-prime variants are claimed integrated. There is no model for how the large-prime cut interacts with orbit quotients (H062, H057 only handle the full-rank threshold).
  - Nearly every cheapest_test is on n=17-31 toys while targets are n=53..131. Only H090/H097/H098/H099 touch n=53+. No idea has a scaling-extrapolation test (a fit over n with a prediction interval at n=131), so toy wins cannot be turned into a distance to rho.
  - No idea reports the matched-rho gap or prices unpriced native work at 0/0.1/1 GAE, apart from H049, H090 and H076. Per section 5a, each should name its rho reference.
  - Hybrid guess strategy for Gröbner solving (how many variables to guess, with the bilinear structure) has only H019/H022/H071. Nothing covers dedicated solvers, such as SAT or XL with the Semaev structure over GF(2) and Frobenius-twisted equations, or specialised F4 linear algebra (H030 only touches the backend).
  - Handbook areas underused: Chapter 8-style zeta/L-function and point-count machinery beyond Weil bounds, Chapter 11.4 factorization (only H034), Chapter 13.4 sparse structured methods (used for PLE only), isogeny/endomorphism ideas outside j=0 (H087 only), and Chapters 16-17 coding/quantum (H091 aside). The Kloosterman/Walsh family (H001,H039,H048) is reused repeatedly with little diversity.
  - Curve-selection has six ideas, three are closed or null (H088,H089,H086), and none addresses GLV/GLS-compatible order policy or composite-order cofactor choice.

- Cross-phase composition is missing: no idea tests that two accepted gains (for example the H007 iota quotient plus the H044 order-6 quotient, or the H045 stop rule plus the H059 core) compose multiplicatively on the same session.

Critic’s top ten for a first measured improvement

id	title	phase	prior art	why
H044	Order-6 automorphism-orbit quotient for j=0 curves, q=1 mod 3	relation-collection	tried-open	Order-6 (negation plus mu_3) quotient on j=0 curves cuts relations and LA about 3x, with a precise ratio falsifier (<2.5) and a log-consistency check. It reuses the existing IC and is a real algorithmic change. H054 and H087 are variants of it. Limited to j=0 prime-field curves and has no binary-curve effect.
H045	Stop collection at orientability threshold; target-only left-kernel extraction	relation-collection	partially-tried	Target-only left-kernel stopping at N_dep/B about 0.92 (m=3) and 0.5 (m=2) is a real cut in relation count. It can be tested by replaying stored relation streams, and the recovered t is verified directly. H049 can share the same replay.
H007	2-torsion translation as extra projectivity in subfield orbit quotient	factor-base	partially-tried	Quotienting by Frobenius plus the 2-torsion translation iota halves unknown classes on even-order binary curves. The

id	title	phase	prior art	why
				falsifier is sharp (class ratio >0.6, or code already quotients), and logs are brute-force checkable at n=18. First check whether 2-torsion saturation already does this.
H026	Gray-code Artin-Schreier enumeration over the hyperplane Tr(s)=Tr(b/r^2)	decomposition	partially-tried	Gray-code Artin-Schreier enumeration over Tr(s)=Tr(b/r^2) returns an identical m=2 relation set at a claimed 8-12x fewer multiplications. Set equality makes it a clean pass or fail on a real hot loop. H024 is the m=3 sibling, and H026's cache-bound caveat is the main risk.
H097	Batch-affine vs projective candidate generation, binary vs prime split	hardware	partially-tried	Batch-affine versus projective addition in GF(2^131) with CLMUL is unmeasured on the binary side, which is the main target class. It is a pure speedup on the collection hot loop with an A/A arm, and H099 supplies the Itoh-Tsujii inversion arm.
H012	Factor base restricted to cofactor image subgroup via character test	factor-base	partially-tried	Restricting the factor base to the cofactor image subgroup raises m=3 success by h (2 for Koblitz

id	title	phase	prior art	why
				a=1, 4 for a=0). It is a mechanism change rather than accounting, with an explicit CI falsifier on yield ratio and a brute-force membership check. The matched rho in H is cheaper by sqrt(h), so the report must carry the net gap.
H059	Closed-form peeling core with max-degree inactivation ordering	relation-filtering	partially-tried	Peeling plus inactivation then block Wiedemann on the core predicts >=1.5x fewer modmuls than the unfiltered matrix. It is testable on saved toy matrices with a falsifier against structured GE plus Wiedemann. H060 is a related variant, and H057 provides the threshold model.
H066	Functional-only Wiedemann reading kernel coordinates from the BM sequence	linear-algebra	partially-tried	Reading kernel coordinates from the BM sequence cuts matvecs from 3N to 2N+O(1), a 33% cut with identical output. It is a deterministic check on 3000x3000 toy matrices, and the unquestioned LA phase is the weakest-covered one.
H041	Trace-form isotropic block	decomposition	partially-tried	Isotropic block pairs kill the

id	title	phase	prior art	why
	pairs giving free linear equations			bilinear part on dim M coordinates and are predicted to lower first-fall degree by >=2 and F4 time about 5x. The falsifier is clear (dim M no larger than random, F4 ratio <1.5). This is the largest potential decomposition-phase gain, and it merges with H038/H027.
H098	Booth/Duval canonical key at n=131 against squaring chains	hardware	partially-tried	Booth/Duval canonical key at n=131 is claimed to give 1.6-3x fewer cycles than min-over-squarings, with identical orbit partition as a hard check. It addresses the key cost in the Frobenius-orbit loop. Merge H063/H092 into the same microbenchmark. Needs a PMU host.

Claim limits

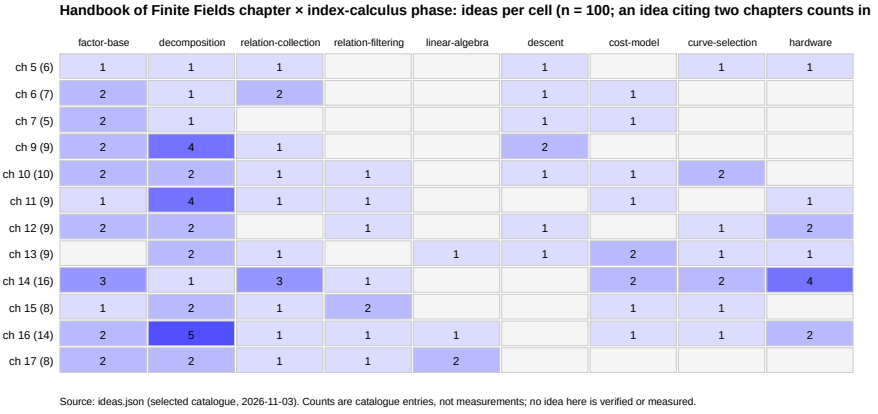
- Every idea is a proposal. The predictions are the authors' preregistered numbers and were not checked against any run. No cost, speedup or rho gap is reported because nothing was measured.
- The prior-art statuses come from a grep over local records by Haiku agents, confirmed by opening at most two files per idea; the atlas match and the handbook knowledge-base catalogue are open checks. A `tried-closed` idea kept in the catalogue is kept because the selector judged the handbook angle new; the critic disputes several (H002, H020, H065, H078, H083, H088, H089, H100).
- The Lean theorems are supporting lemmas about abstract definitions; each record's `missing_bridge` names the curve or field fact its `Prop` assumes. No theorem models the elliptic-curve group law, and a stated `Hnnn_claim` is not evidence that it holds.

- The ToC photographs were transcribed by the miners (`pages.json`); section numbers may carry reading errors from rotated pages and should be checked against the book before citation.

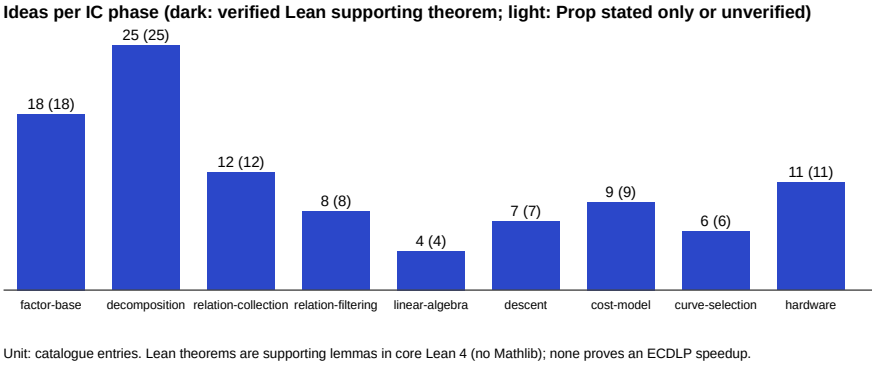
Next step

Merge the duplicate chains, drop the relabellings, and take the critic's top ten through the lab's normal route: an `ecbench-bounds` challenge spec per idea on a toy curve with a matched rho reference, an A/A arm and the §5a GAE band, scheduled through the experiment director. H044/H007 (group-structure quotients) and H045 (early stopping) are the cheapest to decide.

Figures



Chapter by IC phase heatmap



Ideas per phase with Lean status

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100 index-calculus ideas from the Handbook of Finite Fields (2026-11-03)

Each row is a proposal: mechanism, falsifiable claim, preregistered prediction, cheapest discriminating test, falsifier, prior-art status in this lab, and its Lean declaration. Nothing in this table is verified, measured or an improvement; a row becomes a success only

when a frozen, replayed measurement lowers cost against the named IC baseline (AGENTS.md, success criterion).

Lean status key: `proved` = a supporting theorem compiled and passed the axiom audit; `stated` = only the claim `Prop` exists; `unverified` = the module failed the mechanical check (see `formal/receipts/`).

factor-base (18)

H001 — Kloosterman/Walsh ranking of affine cosets for the factor base

- **Handbook**: 17.2.1 Mutually unbiased bases · **Prior art**: partially-tried · **Lean**: proved (`H001_claim`; theorems: `H001_partition_sum`, `H001_no_gain_without_variance`)
- **Claim**: Valid-point counts across the 2^(n-m) cosets of a fixed V exceed binomial spread; a dual-sum (Walsh) predictor ranks cosets with Spearman >= 0.6, so best-of-K gives >= 1.05x relations per factor-base element. A13 measured coset choice as near-null, so this is a predictor test only.
- **Prediction**: Spread 1.0-1.5x binomial SD; predictor rho 0.6-0.9; best-of-64 gain 1.03-1.10x, shrinking with n.
- **Cheapest test**: n=17, m=8: all 512 cosets; count valid points and S3 relations over 2000 targets; correlate with Walsh prediction; 5 random b.
- **Falsifier**: Spearman < 0.3, or best-of-64 gain < 1.02x with bootstrap CI containing 1.
- **Lean gap (assumed, not proved)**: Walsh/Kloosterman dual-sum predictor correlation (Spearman >= 0.6) with coset valid-point counts and the resulting best-of-64 gain; claim quantifies over observation records.

H002 — Cyclic-shift window factor base: Frobenius merge factor about 2

- **Handbook**: 17.1.6.3 Linear polynomial dynamical systems · **Prior art**: tried-closed · **Lean**: proved (`H002_claim`; theorems: `H002_stay_count`, `H002_claim_holds`)
- **Claim**: New angle only on the closed census: for a window V=span{x,…,$x^{(2}$(m-1))} of dim about n/2, Frobenius classes number |V|/c with c in [1.8,3] independent of n, so orbit quotient shrinks columns about 2x, not n x. The existence obstruction itself is closed.
- **Prediction**: Mean class size about 3.0; classes/|V| about 0.5 (0.33-0.56); LA B^2 term falls at most about 4x.
- **Cheapest test**: n=19,29 (also 11,13,37): enumerate window elements, compute classes by repeated squaring, report class-size distribution.
- **Falsifier**: Column reduction >= 4 for the window at n=29 or 37.
- **Lean gap (assumed, not proved)**: That Frobenius class sizes of the window span{x,…,$x^{(2}$(m-1))} in GF(2^n) are at most 3 and sum to |V| (hypothesis of the claim); the shift action as a bit shift on window coordinates.

H003 — Kneser/Freiman-extremal subspaces with small product span

- **Handbook**: 16.7.2 Polynomial multiplication, 16.7.3 Matrix multiplication structure · **Prior art**: partially-tried · **Lean**: proved (`H003_claim`; theorems: `H003_ap_sums`, `H003_claim_holds`, `H003_ap_le_generic`)

- **Claim**: Among d-dim subspaces of GF(2^n), n prime, dim(V*V)=2d-1 holds only for u*span{1,w,..,w^(d-1)}; random subspaces give min(n,d(d+1)/2). Product-span size controls the linearised-decision candidate dimension.
- **Prediction**: n=23,d=7: random V gives dim 23; interval V gives 13; no non-AP subspace in 1000 samples drops below 2d-1.
- **Cheapest test**: n=17..23, d=5..8: sample subspaces, Gaussian-eliminate pairwise products, histogram dim(V*V).
- **Falsifier**: A non-AP subspace with dim(V*V)<2d-1 at prime n, or random subspaces below the generic value at >1%.
- **Lean gap (assumed, not proved)**: Kneser/Freiman lower bound dim(V*V) >= 2d-1 and the equality characterisation (V is u*span{1,w,..,w^(d-1)}) in GF(2^n), n prime; both are hypotheses. The AP sumset count is proved on exponents only.

H004 — Fano-factor selection of equal-size subspace factor bases

- **Handbook**: 16.2.1 Basic concepts of stream ciphers · **Prior art**: partially-tried · **Lean**: proved (`H004_claim`; theorems: `H004_cost_divides`, `H004_cost_antitone`, `H004_cost_improves`)
- **Claim**: Solution count N(V) over random k-dim subspaces is over-dispersed against a symmetric null, so best-of-M selection gives >= 1.2x relations per candidate at fixed |V|.
- **Prediction**: Fano factor exceeds null by >1.5; best of 300 gives 1.2-1.6x; random-b negative control 1.0+-0.05.
- **Cheapest test**: n=19,k=7,m=3: 300 random V, brute-force N(V); compare dispersion with null and Frobenius-invariant positive control.
- **Falsifier**: Dispersion within 0.9-1.1 of null, or best-of-M gain <1.05 after multiple-comparison correction.
- **Lean gap (assumed, not proved)**: Over-dispersion of the solution count N(V) over random k-dim subspaces of GF(2^n) (Fano factor vs symmetric null) and the measured best-of-M gain >= 1.2x; proved part is only the cost arithmetic.

H005 — Trace-code rank-profile criterion for factor-base subspace shape

- **Handbook**: 15.1.2.4 Subfield and trace codes, 15.1.2.3 Reed-Muller codes · **Prior art**: partially-tried · **Lean**: proved (`H005_claim`; theorems: `H005_sum_counts`, `H005_best_ge`)
- **Claim**: At fixed dimension, dispersion index of decomposition counts over targets differs across V shapes (prefix, contains 1, trace-zero, random); the low-rank mass of Dickson forms B_gamma predicts which shape maximises P(N0>=1).
- **Prediction**: Random V index within 5% of 1; prefix and 1-containing V 1.15-1.6; P(N0>=1) differs >= 0.03 between shapes.
- **Cheapest test**: Exact N0(r) for all r at n=17..23 per shape via Artin-Schreier enumeration; 20 random V per shape; subfield-like degenerate positive control.
- **Falsifier**: All shapes within 5% dispersion and 0.01 in P(N0>=1), or predictor Spearman < 0.5.
- **Lean gap (assumed, not proved)**: Dickson-form low-rank (trace-code) predictor linking V shape to P(N0>=1) and the exact Artin-Schreier N0(r) enumeration; double counting is proved for an abstract pair-to-r map, not the field.

H006 — Cyclotomic-coset factor base x^s in a Frobenius orbit

- **Handbook**: 14.7.1 Association schemes, 14.6.5 Cyclotomic classes · **Prior art**: partially-tried · **Lean**: proved (`H006_claim`; theorems: `H006_stable`, `H006_instance`)
- **Claim**: F={x: x^s in C}, C a Frobenius orbit in mu_M, is Frobenius-stable for any n with |F|=s*n, no table needed; yield equals a random stable set within 10%. Divisors of 2^n-1 near 2^(n/2) limit availability.
- **Prediction**: Yield ratio 0.9-1.1; membership 1.0-1.5x a hash lookup at |F| about 2^28; n=29, s=1103 gives |F|=31987.
- **Cheapest test**: n=29, s=1103: compare m=2 yield and columns against random orbit union and Hamming ball; list divisors of 2^n-1 near 2^(n/2) for n in 53..131.
- **Falsifier**: Yield ratio outside [0.8,1.25], or no divisor within 8x of 2^(n/2) for most n.
- **Lean gap (assumed, not proved)**: GF(2^n)* as Z/(2^n-1) via a primitive element, x^s as multiplication by s, ord_M(2)=n for M | 2^n-1, yield-vs-random-stable-set ratio and the existence of divisors of 2^n-1 near 2^(n/2); the |F| = s*n count is checked only by decide for N=63, s=3, n=6.

H007 — 2-torsion translation as extra projectivity in subfield orbit quotient

- **Handbook**: 14.4.2 Collineations and coordinate frames · **Prior art**: partially-tried · **Lean**: proved (`H007_claim`; theorems: `H007_monotone`, `H007_half`, `H007_claim_holds`, `H007_cost_improves`)
- **Claim**: For even-order binary curves with iota-closed subfield base (sqrt b in V), quotienting by <Frobenius, iota> instead of halves unknown classes and relations needed; 2-torsion saturation may already do this.
- **Prediction**: Class ratio 0.50+-0.05; relations to full rank 0.50+-0.10; dense LA x0.25-0.5.
- **Cheapest test**: $y^{2+xy=x}3+x^2+1$ over GF(2^18), V=GF(2^6), m=3: build relation matrices on pi-classes vs (pi,iota)-classes, verify logs by brute force.
- **Falsifier**: Class ratio >0.6, a mod-l inconsistent relation, or code already quotients by iota.
- **Lean gap (assumed, not proved)**: That the 2-torsion translation iota acts freely and commutes with Frobenius on the iota-closed subfield base (sqrt b in V), giving orbit size 2d, plus mod-l consistency of relations transferred to the quotient DLP instance.

H008 — Exponential-sum bias of S3 triple count T(V) across subspaces

- **Handbook**: 12.5.3 Bounds for rational places, 12.7.3 Case of curves · **Prior art**: partially-tried · **Lean**: proved (`H008_claim`; theorems: `H008_le_cube`, `H008_cost_antitone`, `H008_perm_invariant`)
- **Claim**: T(V), the count of S3 solutions in V^3, varies beyond clumped-Poisson across V; offline search beats a random V by more than noise, while structured families (polynomial basis, trace-kernel) stay within 5% of random.
- **Prediction**: std/mean of T 10-25%; best of 1000 gives 1.2-1.5x mean; structured families within 5%.
- **Cheapest test**: n=20,l=7: T(V) for 2000 random V and 6 structured families; dispersion vs Poisson with A/A split.
- **Falsifier**: Dispersion within 1.1 of clumped-Poisson and no structured family beyond 5%.

- **Lean gap (assumed, not proved)**: Exponential-sum/Weil-type bias of the S3 triple count T(V) across subspaces and the best-of-1000 gain of 1.2-1.5x; the S3 relation is an abstract Bool predicate here. Permutation invariance is proved only for a single-index count, not the full triple count.

H009 — Artin-Schreier splitting density of cosets of V

- **Handbook**: 12.1.3 Extensions of function fields · **Prior art**: partially-tried · **Lean**: proved (`H009_claim`; theorems: `H009_tuples_length`, `H009_good_count`, `H009_monotone`)
- **Claim**: Only about half of V lies on E (Tr(v+a+b/v^2)=0); the best coset has density 0.5+O(sqrt(k(n-k))2^(-k/2)), raising useful fraction by (rho/0.5)^m, visible at small k and vanishing at cryptographic size. Koblitz trace-zero policy already forces density 1.
- **Prediction**: n=24,k=8,m=3: best of 10^4 cosets rho 0.60-0.70, gain 1.7-2.7x; n=48,k=16 gain <1.15x.
- **Cheapest test**: n=20..30: E-density of 10^4 random cosets; fit best density to Chernoff prediction; S3 solves on best vs median V.
- **Falsifier**: Best-of-10^4 density within 0.02 of 0.5 at n=24,k=8, or gain <1.3x useful relations per call.
- **Lean gap (assumed, not proved)**: Artin-Schreier splitting criterion Tr(v+a+b/v^2)=0 defining the on-curve predicate, the Chernoff best-coset density 0.5+O(sqrt(k(n-k))2^(-k/2)), and Koblitz trace-zero policy; the strict-improvement conjunct of the claim is stated but not proved.

H010 — Legendre-sequence bias selection of affine cosets (odd p and binary)

- **Handbook**: 10.4.4.3 Legendre sequence and related bit sequences · **Prior art**: partially-tried · **Lean**: proved (`H010_claim`; theorems: `H010_claim_holds`, `H010_best_monotone`, `H010_best_ge`)
- **Claim**: Best-of-T coset selection by lift fraction raises it from 1/2 to 1/2+0.35*sqrt(ln T/|V|); gain is a toy-scale effect shrinking like 1/sqrt|V|.
- **Prediction**: |V|=$2^{10,T=2}$10: 2.6-3% gain in B per enumerated x; under 0.1% at |V|>=2^20.
- **Cheapest test**: GF(2^17) and p about 2^20: lift fraction over 2000 cosets of a dim-10 subspace vs Gaussian and random-coset nulls, with A/A.
- **Falsifier**: Max over T cosets within one standard error of the iid-binomial null, or realised gain <0.5% at |V|=2^10.
- **Lean gap (assumed, not proved)**: That the +-1 characters are Legendre/quadratic-character values of the lift predicate on cosets of V in GF(2^n) or F_p, and the 0.35*sqrt(ln T/|V|) extreme-value gain of best-of-T selection; only the integer identity 2lifts = |V| + sum chi and max-monotonicity are proved.

H011 — Primitive-polynomial LFSR enumeration of factor-base abscissae

- **Handbook**: 11.1.1.3 Primitive elements, 11.1.3.2 Modular reduction in extension fields · **Prior art**: untried · **Lean**: proved (`H011_claim`; theorems: `H011_pow_add`, `H011_distinct_lt`, `H011_distinct`, `H011_enum_cost_ratio`)
- **Claim**: Factor base {t^i}, t primitive, is enumerated with one shift plus <=hw(f)-1 XORs, >=10x cheaper than random-x sampling, with relation yield equal to a

random-x base within 10%.
- **Prediction**: Enumeration 5-15x cheaper in bit ops; yield within +-5%; end-to-end gain <2%.
- **Cheapest test**: GF(2^n), n=19,23,29: B=2^10 for both bases, 10^5 targets each, matched seeds, A/A arm, count bit ops.
- **Falsifier**: Yield differs >10%, or enumeration saving <3x in measured cycles.
- **Lean gap (assumed, not proved)**: That t is primitive in GF(2^n) (pow k != 1 for 0<k<2^n-1), that the field multiplication is associative with cancellation, and that LFSR step cost is hw(f) bit ops. The 10x saving and the yield match are not proved.

H012 — Factor base restricted to cofactor image subgroup via character test

- **Handbook**: 10.1.4.3 Characteristic functions, 10.1.4.4 Gauss sums · **Prior art**: partially-tried · **Lean**: proved (`H012_claim`; theorems: `h012_pigeon`, `H012_pair_fibre_average`, `H012_pair_sum_mem`)
- **Claim**: Drawing F only from H=hE(F_q) (Tr(x)=Tr(a) for h=2, half-trace test for h=4) raises per-tuple m=3 decomposition success by h at fixed |F|; matched rho in H is cheaper by sqrt(h), so net gap closes at most sqrt(h).
- **Prediction**: Yield ratio 2.0+-0.15 (h=2), 4.0+-0.4 (h=4) on GF(2^17..21).
- **Cheapest test**: GF(2^19), a=0 and a=1 Koblitz: filtered vs unfiltered F (|F|=2^6), 2000 targets in H; check halving test against brute-force 2E membership.
- **Falsifier**: Yield ratio 95% CI upper bound <1.2 for h=4, or halving test disagrees with brute force on any point.
- **Lean gap (assumed, not proved)**: That H=hE(F_q) has size N/h and is closed under addition, and that the character or half-trace test decides membership. Pair sums are modelled as a list of labels below M.

H013 — Valid-point fraction of Frobenius-invariant subspaces (orbit-level variance)

- **Handbook**: 9.1.2 Walsh transform, 9.1.1.2 Trace representation · **Prior art**: partially-tried · **Lean**: proved (`H013_claim`; theorems: `H013_valid_eq_repsum`, `H013_claim_holds`, `H013_valid_le_total`, `h013_filter_const`)
- **Claim**: For invariant V the point-existence indicator is orbit-constant, so valid fraction has spread about 0.5*sqrt(n/|V|); best-of-K invariant subspaces give yield (1+2eps)^m above median. Moment identities caution that per-curve spectra are invariant.
- **Prediction**: n=23,d=8..9,m=3: spread 0.12-0.16; best of 2000 fraction >=0.75; yield >=2.5x median; generic V <=1.5x.
- **Cheapest test**: 2000 random invariant subspaces at n=23,d about 8: valid fraction vs relations on 500 targets; regress on fraction^3.
- **Falsifier**: Spread equals iid-coin spread, yield ratio <1.3x, or R^2 <0.5.
- **Lean gap (assumed, not proved)**: That the point-existence indicator is constant on Frobenius orbits of V, that the relation count is at most validCount^m, and the spread of order 0.5*sqrt(n/|V|).

H014 — Swap hill-climbing on exact S3 yield N(V)

- **Handbook**: 9.1.3 Parameters of Boolean functions, 9.1.1.1 Algebraic normal form · **Prior art**: partially-tried · **Lean**: proved (`H014_claim`; theorems: `H014_fibre_le_two`, `H014_total_le`, `h014_two_bound`, `h014_flat_len`)

- **Claim**: N(V) is overdispersed across subspaces; swap hill-climbing finds V with several times median yield in a few hundred evaluations, transferring to fresh targets.
- **Prediction**: Var/Mean >=3 at n=23,d=8; climbed V >=2x median, >=1.8x on fresh targets; <=1.5x at n=31.
- **Cheapest test**: Brute-force N(V) for 2000 random V at n=19..23; 500-step swap climbs from 20 starts; retest on 2000 fresh targets.
- **Falsifier**: Dispersion <1.5, or climbed V <1.2x median on fresh targets.
- **Lean gap (assumed, not proved)**: That S3 solutions in x3 are the fibre of an xor-additive map such as x^2+x with kernel {0,e}, and that V is duplicate-free. Overdispersion of N(V) and hill-climb transfer are not proved.

H015 — Best-of-T scaling/translate of V by Kloosterman liftable bias

- **Handbook**: 7.1.4 Artin-Schreier hypersurfaces, 6.3.5 Incomplete character sums · **Prior art**: partially-tried · **Lean**: proved (`H015_claim`; theorems: `H015_sum_le_len_mul_max`, `H015_rel_antitone`, `H015_cost_improves`, `h015_le_max`)
- **Claim**: Solutions need all x_i liftable, so yield scales as (2*liftFrac)^m; best of T scalings lambda*V (or translates) raises liftFrac by about sqrt(2 ln T/2^d)/2 at unchanged solve cost. Prime-degree lift repulsion is a closed-leaning negative.
- **Prediction**: GF($2^{29),d=10,T=2}$10: liftFrac about 0.55, m=3 yield ratio about 1.33 (1.2-1.5); at d=20 only about 1.03.
- **Cheapest test**: n=29,d=10: 1000 random V vs best-of-1024; count S3 solutions over 50 x_R, paired.
- **Falsifier**: Yield ratio <1.1, or fails to track (liftFrac ratio)^m within 15%.
- **Lean gap (assumed, not proved)**: That yield scales as (2*liftFrac)^m, that the liftCount list comes from Kloosterman-liftable x over the scalings lambda*V, and that the gain is about sqrt(2 ln T/2^d)/2. The yield ratio c is a natural-number stand-in.

H016 — Affine-hull dimension of S3(V^3) as subspace selection criterion

- **Handbook**: 6.4.2 Sum-product estimate · **Prior art**: partially-tried · **Lean**: proved (`H016_claim`; theorems: `H016_antitone`, `H016_generic_at_full`, `H016_hull_at_least_generic`, `H016_claim_holds`)
- **Claim**: Solution count over V^2 is about |V|^2*2^(-w(V))*[c0 in hull], w the hull dimension (about dim span(V*V)); K-structured V has w<n and high yield for hull-compatible x_R but risks rank collapse in E(K).
- **Prediction**: n=24,k=6: V=K+K*theta, w<=18, yield about 64x generic when c0 in hull; average over x_R unchanged unless hull is steerable.
- **Cheapest test**: n=24,k=6: random vs K-structured V; compute w by elimination, count solutions on 200 (V,x_R) draws, correlate log2 count with n-w; check relation rank mod ell.
- **Falsifier**: Corr <0.8, average yield equal to generic within 5% with no steering, or rank <90% of columns.
- **Lean gap (assumed, not proved)**: That the solution count is |V|^2*2^(-w) times [c0 in hull], that hullDim is about dim span(V*V), and that rank does not collapse in E(K). The formula for h016Rel is assumed.

H017 — Legendre-biased multiplicative coset selection for prime-field factor bases

- **Handbook**: 7.1.5 Kummer hypersurfaces, 7.3.2 Jacobi sums · **Prior art**: partially-tried · **Lean**: proved (`H017_claim`; theorems: `H017_agree_ge_half`, `H017_cost_antitone`, `H017_claim_holds`)
- **Claim**: x3 in F_p iff chi(f(x1))=chi(f(x2)), so yield from coset gH goes as (2 liftFrac)^m; best of T cosets gains (1+sqrt(2 ln T/B))^3, vanishing at production size. Window families are closed; coset selection is untested.
- **Prediction**: p about $2^{40,B=2}$10,T=2^10: liftFrac about 0.558, gain about 1.39; gain-1 scales as B^(-1/2).
- **Cheapest test**: p about 2^40 curve; 1000 random cosets vs best-of-1024; count S3 solutions by x1,x2 enumeration; scaling over B=$2^{8..2}$14.
- **Falsifier**: Gain <1.1 at B=2^10, or log-log slope of gain-1 outside -0.5+-0.15.
- **Lean gap (assumed, not proved)**: That x3 is in F_p iff chi(f(x1))=chi(f(x2)), the Legendre bias of the coset gH, and the gain (1+sqrt(2 ln T/B))^3. The best-of-T statement reuses the H015 abstract max-versus-average lemma.

H018 — Rotation-closed Hamming-ball necklace factor base for primitive-root n

- **Handbook**: 5.2.1 Normal bases, 5.3.1 Low complexity normal bases, 14.9.2 Weights of multiples · **Prior art**: partially-tried · **Lean**: proved (`H018_claim`; theorems: `H018_wt_rot`, `H018_ball_closed`, `h018_fixed_const`, `h018_wt_const`, `H018_fixed_weight`)
- **Claim**: Where no invariant subspace exists, W_w={wt_ONB<=w} is Frobenius-closed with necklace storage |W_w|/n and popcount membership; yield equals a random closed set within 5%. Encyclopedia C6 rejected it as storage-only, so the claim is a cost, not yield, gain.
- **Prediction**: Yield ratio 0.95-1.05; necklace count exact by Burnside; membership >=1.3x faster than hash at 2^28; end-to-end <=1.1x.
- **Cheapest test**: n=29, |W_w| about 2^{12-2}17: m=2 collection on a Koblitz toy vs random closed set and non-stable set of equal size; microbenchmark membership.
- **Falsifier**: Yield ratio outside [0.9,1.1], orbit count off Burnside, or popcount not faster than hashing.
- **Lean gap (assumed, not proved)**: That Frobenius acts as cyclic rotation in the ONB. Rot-by-1 fixed points are proved to be 0 or all-ones for any n. The prime-n orbit structure of rot^k (exact Burnside necklace count), the popcount speedup and the yield match are not proved.
- Merged from pool keys: P5-1

decomposition (25)

H019 — Groebner-fan bi-weight order for the bilinear S3 subspace system

- **Handbook**: 17.1.4.2 Groebner fan parameter estimation · **Prior art**: partially-tried · **Lean**: proved (`H019_claim`; theorems: `H019_zero_left`, `H019_kernel`, `H019_len_decomp`, `H019_toy_count`)

- **Claim**: On bilinear S3 with |V|=|W|=2^m, m=8..12, the best fan-selected bi-weight order cuts the largest F4 matrix >=2x vs grevlex, yet F4 stays above the 2^m lookup baseline.
- **Prediction**: Matrix ratio 0.3-0.7, shrinking with m; F4 time >=10x the lookup baseline at m>=10.
- **Cheapest test**: n=20, 50 targets: msolve/F4 under grevlex, ($1^{10,0}10$) refined by grevlex, and 5 random fan weights; compare matrix size, dreg, time.
- **Falsifier**: Best bi-weight matrix >=0.8x grevlex, or loses a paired sign test (p<0.05).
- **Lean gap (assumed, not proved)**: That S3 restricted to V x W is xor-bilinear (additivity of squaring in char 2). Groebner-fan matrix sizes, F4 cost and the lookup-baseline comparison are not modelled.

H020 — Canalyzing-depth pre-sieve for the post-trace decomposition predicate

- **Handbook**: 17.1.4.1 Minimal-sets algorithm, 17.1.6.1 Nested canalyzing functions · **Prior art**: untried · **Lean**: proved (`H020_claim`; theorems: `H020_presieve_sound`, `H020_detset_mono`, `H020_canal_is_detset`, `H020_parity3_no_canal`, `H020_and3_canal`, `H020_claim_holds`)
- **Claim**: P(a)=[x2 in W] and the S3 bit components have canalyzing depth 0 for >=99% of targets, minimal determining set >=0.9m, so no m/4-bit pre-sieve discards >5% beyond the trace test. Expected negative; closes the route.
- **Prediction**: Depth 0 in >99%; determining set 0.9-1.0m; pre-sieve discards 0-3%.
- **Cheapest test**: n=18,m=9, 500 targets: enumerate P, brute-force subsets up to 3 for canalyzing variables; random-function negative control.
- **Falsifier**: Determining set <=m/4 for >5% of targets, or pre-sieve discards >=10% losslessly.
- **Lean gap (assumed, not proved)**: That the post-trace decomposition predicate P(a) and the S3 bit components behave like the toy predicates, with canalyzing depth 0 for at least 99% of targets and determining set at least 0.9m. Only soundness and monotonicity of the pre-sieve and two 3-bit instance checks are proved.

H021 — Frobenius-linear decision test for 2-point decomposition on interval bases

- **Handbook**: 16.7.2 Polynomial operations arithmetic, 16.3.3.12 Linearization · **Prior art**: partially-tried · **Lean**: proved (`H021_claim`; theorems: `H021_sound`, `H021_relCost_le`, `H021_claim_proved`, `H021_instance_n31_d9`)
- **Claim**: For V=span{1,..,t^(d-1)}, 3d-1<=n, S3 is F2-linear in bits of (pi,s) with pi reduction-free; deciding x(R)=x1+x2 costs one n x (3d-1) elimination, not 2^d, returning identical decompositions.
- **Prediction**: Per-relation cost 2^(n-2d)*n^3/64; total about 2^(2n/3) at d=n/3; >=100x speedup at n=31,d=9.
- **Cheapest test**: Koblitz n=31 (also 23,37), d=9: 2^16 targets, compare decomposition sets and time vs enumeration; fit log2 cost slope vs n.
- **Falsifier**: Any missed decomposition, speedup <10x at n=31,d=9, or slope not within 20% of 1/3.
- **Lean gap (assumed, not proved)**: S3 is F2-linear in the bits of (pi,s) on interval bases with 3d-1<=n, so the n x (3d-1) elimination decides exactly h021Decomposes; taken as the hypothesis hlin. The 2^(n-2d)*n^3/64 cost formula and the speedup are not derived.

H022 — Block-multigraded Macaulay matrices for Weil-restricted S_{m+1}

- **Handbook**: 16.3.3.12 Direct attacks with polynomial solvers, 16.3.3.8 Rank structure · **Prior art**: partially-tried · **Lean**: proved (`H022_claim`; theorems: `h022Mono_pos`, `h022Mono_le_succ`, `h022Mono_mono`, `H022_bitDeg_le_two`, `H022_claim_proved`)
- **Claim**: Bit-degree popcount(e) per block makes the system multihomogeneous; at the first-fall multidegree the multigraded Macaulay matrix has >=10x fewer columns than total degree at m=3. m=2 bidegree was measured column-heavy.
- **Prediction**: Column ratio >=10 at d=7,n=21, growing with d; first fall at multidegree (1,1,1) or (2,1,1).
- **Cheapest test**: Build sparse F2 Macaulay matrices for m=3, n=21, d=7 in both gradings to degree 3; compare columns and rank at recovery.
- **Falsifier**: Column ratio <2 at first fall, or multigraded rank fails to recover the planted decomposition at smaller size.
- **Lean gap (assumed, not proved)**: Macaulay columns are modelled by squarefree monomial counts, not by the actual sparse matrix. The first-fall degree of the total-degree system (assumed at least 5) and the first-fall multidegree (1,1,1) are not derived. Matrix rank at recovery is not modelled.

H023 — One-bit trace criterion as a dead-target sieve for 2-decompositions

- **Handbook**: 16.3.3.1 Linearization equations, 16.3.2.4 Frobenius-additive structure, 9.1.1.2 Trace representation · **Prior art**: partially-tried · **Lean**: proved (`H023_claim`; theorems: `H023_filter_preserved`, `H023_live_le`, `H023_partition`, `H023_claim_proved`, `H023_strict_improvement`)
- **Claim**: For V in ker Tr, targets with Tr(b/z^2)=1 (equivalently Tr(x_R)!=(m-1)Tr(a)) have no V-decomposition; live targets carry 2x yield, halving solves per relation. Leads record this as closed (<=0.2 bits, degree-2 fall); remaining angle is the m=2 loop savings.
- **Prediction**: Dead fraction 50+-2%, zero counterexamples, live mean 2.0x, additions ratio 0.50+-0.03.
- **Cheapest test**: n=17,19, b=1 and random b, k=8 subspaces inside/outside ker Tr: enumerate x1, tabulate counts against Tr(b/z^2).
- **Falsifier**: Any decomposition of a Tr=1 target for V in ker Tr, dead fraction outside 0.48-0.52, or live mean not near 2x.
- **Lean gap (assumed, not proved)**: The hypothesis h_dead: for V in ker Tr, Tr(x_R) != (m-1)Tr(a) implies no V-decomposition. The 50 percent dead fraction and the 2x live yield are not derived.
- Merged from pool keys: P12-0

H024 — Lifted (s,p) enumeration with half-trace bit-GEMM tables

- **Handbook**: 16.3.2.5 Oil and vinegar, 16.3.3.2 Critical bilinear relations · **Prior art**: partially-tried · **Lean**: proved (`H024_claim`; theorems: `h024_decode_lifted`, `H024_claim_proved`, `H024_ratio`, `H024_xor_cancel`)
- **Claim**: For fixed z, S3 is affine in (s,p); enumerating s in V with precomputed half-trace and 1/s^2 tables uses <=0.6 field multiplications and no inversion per candidate vs >=4, returning the identical relation set.

- **Prediction**: Identical multiset; 0.5+-0.1 mults per candidate; 3-6x faster at n=45,k=15, <2x once tables exceed L2.
- **Cheapest test**: n=33 (m=2,k=16) and n=45 (m=3,k=15): count mults and time both loops over 1000 targets; compare relation sets.
- **Falsifier**: Relation sets differ, gain <1.5x at k<=18, or table traffic erases the saving.
- **Lean gap (assumed, not proved)**: The multiplication counts (at most 0.6 against at least 4 per candidate) are constants, not derived from half-trace and 1/s^2 table lookups. That S3 is affine in (s,p) for fixed z and that a lifted witness corresponds to Dec is not modelled. Only the pure (s,x1) re-indexing bijection on XOR-closed lists is proved, so relation-set equality holds up to the swap.

H025 — Optimal normal basis coordinates to minimise monomials of Weil-restricted S3

- **Handbook**: 16.3.1.1 Bipolar construction, 16.3.1.3 Isomorphism of polynomials · **Prior art**: partially-tried · **Lean**: proved (`H025_claim`; theorems: `H025_count_invariant`, `H025_cost_scaling`, `H025_claim_proved`)
- **Claim**: Choosing S,T as consecutive ONB elements gives >=10x fewer monomials and >=3x sparser Macaulay matrices than a random basis pair at equal degree of regularity, lowering F4 time.
- **Prediction**: Monomial ratio about n/2 (10-30x at n 30-60); F4 time 2-5x; dreg unchanged.
- **Cheapest test**: n=29 (type II) and n=28 (type I): k=9 subspace S3 in random vs ONB basis; count ANF monomials; run F4 with fixed hybrid guesses.
- **Falsifier**: Monomial ratio <2 after restriction to V, or F4 time not improved despite density.
- **Lean gap (assumed, not proved)**: The ONB basis change is modelled only as an abstract left-invertible map f with left inverse g on Nat points. The monomial density ratio (monos' * c <= monos), the F4 cost proportional to monomials * columns, and the dreg-unchanged claim are hypotheses or model choices.

H026 — Gray-code Artin-Schreier enumeration over the hyperplane Tr(s)=Tr(b/r^2)

- **Handbook**: 15.1.2.4 Trace codes, 15.1.2.1 Standard array decoding, 15.1.2.3 Reed-Muller codes · **Prior art**: partially-tried · **Lean**: proved (`H026_claim`; theorems: `H026_claim_proved`, `H026_identity_gf8`, `H026_iff_gf8`, `H026_gray_one_bit`, `H026_hyperplane_half`)
- **Claim**: With s=x1+x2, w^{2+rw+r}2s^2+b=0 is solvable iff Tr(s)=Tr(b/r^2); Gray-code walk over V cap hyperplane finds all m=2 decompositions with >=5x fewer multiplications per relation than hash-lookup of x(R-P), same relation set.
- **Prediction**: Multiplication ratio 8-12x, wall-clock 3-6x (1/s^2 table memory-bound).
- **Cheapest test**: n=31,l=15, 10^4 targets: count multiplications and cache-missing reads in both; check set equality; repeat n=41,l=20.
- **Falsifier**: Relation sets differ, multiplication ratio <2, or wall-clock ratio <1.
- **Lean gap (assumed, not proved)**: The Artin-Schreier identity is checked by decide only over GF(8), not for general n such as 31 or 41. Solvability iff Tr(s)=Tr(b/r^2) (hypothesis hsolv), the trace map itself, and the multiplication and cache-miss counts are not modelled. The Gray walk is stated only as V.filter hyp.

H027 — Rank profile of Weil-restricted S3 components as F4 degree predictor (invariant vs random V)

- **Handbook**: 15.1.2.3 Reed-Muller codes, 15.1.2.4 Trace codes, 15.1.3 Cyclic codes · **Prior art**: partially-tried · **Lean**: proved (`H027_claim`; theorems: `H027_claim_proved`, `h027Ins_length`, `h027Insert_length_le`, `h027Rank_fold_le`, `H027_rank_le_rows`, `H027_rank_instances_n7`)
- **Claim**: For m=2, n=23..31, l=n/2, Frobenius-invariant V gives F4 solving degree >=1 below random V of equal dimension, predicted by the dimension of the low-rank component subspace.
- **Prediction**: Degree drops 4->3 in >=60% at n=31,l=15; low-rank dimension >=3 higher; time 1.5-3x lower.
- **Cheapest test**: n=31 and n=21: 50 targets, rank profile of n component forms, F4 (msolve/M4GB) degree and time; invariant V vs 20 random V.
- **Falsifier**: Paired bootstrap CI of degree difference contains 0, or Spearman(low-rank dim, degree) <0.4.
- **Lean gap (assumed, not proved)**: The mechanism hypothesis (low-rank-dimension excess of at least 3 implies an F4 degree drop of at least 1) and the 60 percent dimension-excess rate are assumptions. The Weil-restricted component forms, F4 degrees and basis-change invariance of rank are not modelled; rank is only bounded by the row count and checked on n=7 examples.

H028 — Determinantal elliptic-normal-curve system keeping y

- **Handbook**: 14.4.5 k-Arcs, 14.4.1 Projective and affine spaces · **Prior art**: partially-tried · **Lean**: proved (`H028_claim`; theorems: `H028_var_balance`, `H028_time_ratio_implies_faster`, `H028_balance_m3_n21`)
- **Claim**: The det[phi_j(P_i)]=0 system on |kO| keeping y, with Weil-restricted x only, has lower F4/F5 degree of regularity and solve time than S_{m+1} at equal V for m=3,4.
- **Prediction**: dreg lower by >=1 at m=3; time ratio 0.3-0.8 for n<=25, vanishing at n>=40.
- **Cheapest test**: E over GF(2^21), dim V=7, m=3: build both systems for 200 targets, run same F4, record dreg, matrix size, time, matching projected solutions.
- **Falsifier**: dreg not lower, median time ratio >=1, or solution sets differ.
- **Lean gap (assumed, not proved)**: The hypothesis hdet (sum of logs zero iff the determinant vanishes on |kO|) is not stated as a theorem. The dreg drop, time ratio and solution-set equality are measured outcomes held in a record, not derived. Only the variable count arithmetic m*(n/m+n) = (m+1)*n is proved.

H029 — Rank-count law as null model for S3 pair matrices

- **Handbook**: 13.2.1 Matrices of specified rank, 13.3.5 Orthogonal groups, characteristic two · **Prior art**: partially-tried · **Lean**: proved (`H029_claim`; theorems: `H029_coset`, `H029_claim_proved`, `H029_fibres_instance`)
- **Claim**: S3 is F2-affine in x1 for fixed (x2,x3), so solution sets are cosets of a kernel; some V (invariant, trace-orthogonal) have rank-deficient M(x2,x3) at >=2x the uniform-matrix rate, raising relations per pair.
- **Prediction**: Random V within 5% of 2^-(n-m+1); best of 200 structured V gives 1.5-4x excess.
- **Cheapest test**: n=19,m=8 and n=23,m=9: 100 subspaces, all 2^16 pairs, bit-packed rank, histogram kernel dimension vs closed-form rank-count law.

- **Falsifier**: All structured V within 10% of null (bootstrap CI contains 1).
- **Lean gap (assumed, not proved)**: That S3 is F2-affine in x1 (the additivity hypothesis hadd) and that a structured V has an at least 2x excess of rank-deficient pair matrices are assumed. The coset result is proved only as an iff, and the cardinality of the kernel coset is not proved. The closed-form rank-count law is not formalised.

H030 — Fill-triggered M4RI/PLE backend for Macaulay matrices (matched-degree test)

- **Handbook**: 13.4.1.1 Tiny finite fields, 13.4.2.2 PLE decomposition · **Prior art**: partially-tried · **Lean**: proved (`H030_claim`; theorems: `H030_m4ri_le_schoolbook`, `H030_trigger_mono`, `H030_instance`)
- **Claim**: Four-Russians PLE is already landed and positive; the open part is a fill trigger (>15% after 30% pivots) on binary Semaev Macaulay matrices n=19..29, giving >=2x fewer bit ops than sparse F4.
- **Prediction**: 2-5x for >=20k x 20k, <1x below 2k x 2k; decomposition total -20-40% at n=29.
- **Cheapest test**: Dump 10 Macaulay matrices at 5k..40k columns, run both backends on the same host, compare time and word-XORs, confirm equal rank.
- **Falsifier**: Speedup <1.2x at 20k x 20k, or fill stays <5%.
- **Lean gap (assumed, not proved)**: The M4RI cost model (p/k blocks, each 2^kc+rc) and the schoolbook model prc are idealised dense counts. Sparse-F4 cost, fill dynamics, the 15 percent after 30 percent trigger actually firing, and the measured 2x speedup at 20k x 20k are not derived. H030_claim itself is a stated measurement prediction and is not proved.

H031 — Halving-trace character appended as an explicit linear equation in the solver

- **Handbook**: 12.2.6 Torsion subgroup, 12.2.5 Twists · **Prior art**: partially-tried · **Lean**: proved (`H031_claim`; theorems: `H031_nsmul_psi`, `H031_combo_psi`, `H031_filter_keeps_valid`, `H031_claim_holds`)
- **Claim**: Adding psi(P)=Tr(x_P)+Tr(a) sum constraint up front and eliminating one bit removes no true decomposition and cuts F4/hybrid time at fixed (n,l,m); fall-degree records suggest F4 gets it at degree 2.
- **Prediction**: Zero decompositions lost; hybrid time ratio 0.5-0.7; dreg change <=1.
- **Cheapest test**: n=31,l=10,m=3, 200 targets: solve with/without the equation (msolve or hybrid guess loop); compare solution sets; brute-force check.
- **Falsifier**: Any decomposition violating the equation, time ratio >0.9, or F4 derives it below d_reg.
- **Lean gap (assumed, not proved)**: That the halving-trace character psi(P)=Tr(x_P)+Tr(a) is a homomorphism to Z/2 on the curve group (the h031Hom record), and that F4/hybrid time ratio and d_reg behave as predicted.

H032 — Affine shift/scale and Montgomery-model scan to minimise S_m monomials

- **Handbook**: 12.3.7 Montgomery curves, 12.3.8 Twisted Edwards, 12.3.5 Short Weierstrass · **Prior art**: untried · **Lean**: proved (`H032_claim`; theorems: `H032_smoothCount_invariant`, `H032_exists_min`, `H032_min_le_origin`, `H032_claim_holds`)

- **Claim**: x->u(x-c) with c,u in the subfield keeps the base intact; nnz(S4) and F4 matrix size vary >25% over (c,u) and 2-isogenous Montgomery models, and short Weierstrass is not the minimum.
- **Prediction**: Montgomery S4 >=25% fewer monomials; min over c <=0.75x nnz at c=0; F4 time 0.8-0.9.
- **Cheapest test**: Build S4 over F_p (p=1009) in each form, scan (c,u), record nnz; msolve on Weil-restricted n=3,m=3 for 5 extremal (c,u).
- **Falsifier**: nnz spread <5%, or F4 time ratio >0.95.
- **Lean gap (assumed, not proved)**: That x->u(x-c) and 2-isogenous Montgomery models are bijections of the decomposition solution set preserving the base (the hypothesis g(f a)=a), and the actual nnz(S4) / F4 time values over the scan.

H033 — Support and Newton-polytope structure of GF(2)-defined Koblitz S_4, S_5

- **Handbook**: 11.5.2.2 Polytope indecomposability, 11.5.2.3 Sparse bivariate Hensel lifting · **Prior art**: partially-tried · **Lean**: proved (`H033_claim`; theorems: `H033_S3_sym`, `H033_S3_nodup`, `H033_S3_box`, `H033_S3_density`, `H033_claim_holds`)
- **Claim**: Over GF(2) in e-coordinates, S_4,S_5 support fills <=30% of the weighted-dense set (Frobenius-closed exponent sets) with Minkowski-indecomposable polytope; density falls with m, giving sparse Macaulay rows.
- **Prediction**: S_4 density 15-40%; S_5 5-25%; polytope indecomposable at m=3,4,5.
- **Cheapest test**: Compute S_3 by hand, S_4,S_5 by resultant over GF(2); count monomials vs weighted-dense; F4 sparse vs dense row storage on S_4 toys.
- **Falsifier**: S_5 density >=60%, or a nontrivial factorization of some S_m.
- **Lean gap (assumed, not proved)**: The actual supports of S_4, S_5 over GF(2) (resultant computation), Frobenius-closure of those exponent sets, the weighted-dense reference set, and Minkowski indecomposability of the Newton polytopes (Indec is an abstract predicate).

H034 — Linearized subspace-polynomial test L_V mod N for root membership

- **Handbook**: 11.3.3 Rabin irreducibility test, 11.4 Univariate factorization · **Prior art**: partially-tried · **Lean**: proved (`H034_claim`; theorems: `H034_mem_of_prod_zero`, `H034_prod_zero_of_mem`, `H034_roots_in_V`, `H034_claim_holds`)
- **Claim**: For degree d<=16 completion polynomial N, L_V mod N = 0 decides all roots in V with about k Frobenius powers, >=2x fewer multiplications than distinct-degree plus equal-degree root finding, identical decisions.
- **Prediction**: n=48,k=16,d=8: multiplication ratio 2.5-4.
- **Cheapest test**: GF(2^n), n=24..48, 10^5 random N of degree 4-16 with 1% planted V-roots; compare decisions and counted multiplications.
- **Falsifier**: Any differing decision, or counted multiplications not lower by 1.5x.
- **Lean gap (assumed, not proved)**: That GF(2^n) is an integral domain (h034Dom), that N | L_V yields L_V vanishing on roots of N, that L_V vanishes exactly on V, and the 1.5x+ multiplication-count advantage over distinct/equal-degree factoring.

H035 — Canonical sign bit Tr(y/x) resolving x-only Semaev solutions

- **Handbook**: 11.1.9.2 Half-trace, 11.1.6.2 Squaring in characteristic two · **Prior art**: partially-tried · **Lean**: proved (`H035_claim`; theorems: `H035_sign_neg`,

`H035_flip_all`, `H035_trials_gt_one`, `H035_claim_holds`)
- **Claim**: For odd n, s(P)=Tr(y/x) is a negation-sign bit and sum s(P_i) equals a closed form in (x1,x2,x3) (one sqrt, trace, inversion), selecting the valid sign class without 2^(m-1) trial additions.
- **Prediction**: Closed form matches brute force exactly; sign-resolution ops fall to about one inversion plus one half-trace; total change <5%.
- **Cheapest test**: 5 toy curves, n in {11,…,19}: all zero-sum triples; compare sum Tr(y_i/x_i) with closed form; time vs brute-force addition for m=3,4.
- **Falsifier**: Any triple disagreeing with the true sign class, or sign resolution <2% of decomposition time.
- **Lean gap (assumed, not proved)**: Field trace additivity and Tr(1)=1 for odd n (h035Tr record), negation being (x,y+x) on the binary curve, and the closed form for sum s(P_i) in (x1,x2,x3).

H036 — Extraneous diagonal factor of S_4(x,x,y,z)

- **Handbook**: 10.5.3 Degree growth · **Prior art**: partially-tried · **Lean**: proved (`H036_claim`; theorems: `H036_deg_factor`, `H036_degree_drop`, `H036_quarter_drop`, `H036_S3_degree2`, `H036_diag_S3`, `H036_claim_holds`)
- **Claim**: S_4(x,x,X3,X4) contains an extraneous square factor from the degenerate P+P=O branch; its squarefree part has lower degree and removing it lowers F4 dreg by >=1 for some (m,l).
- **Prediction**: Squarefree degree about 25% lower; F4 time -20-50% on diagonal subsystem; no effect off-diagonal.
- **Cheapest test**: Singular over GF(2^k), k=7..13: build S_3,S_4, substitute x1=x2, factor; F4 on diagonal Weil-restricted systems with/without the factor.
- **Falsifier**: Substituted polynomial squarefree of full degree, or time and dreg unchanged within A/A.
- **Lean gap (assumed, not proved)**: That S_4(x,x,y,z) actually factors as f^2*g with deg f>0 from the degenerate P+P=O branch (hypothesis s = f*g), and the F4 d_reg/time effect of removing the factor.

H037 — Sparse relative defining polynomial for Weil restriction on composite n

- **Handbook**: 11.1.3.2 Modular reduction, 11.1.1.2 Extension field generation · **Prior art**: partially-tried · **Lean**: proved (`H037_claim`; theorems: `H037_dreg_invariant`, `H037_time_le`, `H037_claim_holds`)
- **Claim**: On composite n=l*m (55,57,91,93,95), a sparse relative defining polynomial lowers F4 time by a constant factor while dreg stays identical to a dense choice.
- **Prediction**: dreg identical; time ratio dense/sparse 1.2-2x at n=57,l=19,m=3; no exponent change.
- **Cheapest test**: n=55 and n=57: restricted S_3 with 5 random irreducibles and the best sparse one; record monomials, dreg, F4 time, randomised order, A/A.
- **Falsifier**: Runtime ratio within A/A noise, or dreg differing across choices.
- **Lean gap (assumed, not proved)**: That the degree-d Macaulay predicates of the dense and sparse relative-polynomial systems coincide (invertible linear substitution invariance), and the measured constants cs<=cd with the 1.2-2x time ratio.

H038 — Dembowski-Ostrom rank spectrum of the fixed-s S_3 system (MinRank screen)

- **Handbook**: 9.5.5 Planar DO polynomials, 10.1.4.2 Nonlinearity · **Prior art**: tried-open · **Lean**: proved (`H038_claim`; theorems: `H038_expansion_diff`, `H038_expansion_mod2`, `H038_exponents_pow2`, `H038_low_rank_exists`, `H038_claim_holds`)
- **Claim**: With x2=x1+s, S_3 is a DO quadratic map; for structured V (subfield-adjacent, Frobenius-stable) the min component rank is >=2 below the random-map null, giving cheaper per-s rank-aware solves than generic F4 hybrid. Open lead, unrun.
- **Prediction**: n=21,d=7: random V matches null (KS p>0.05); structured V min rank lower by >=2 in >=30% of cases.
- **Cheapest test**: n=21,d=7: all 2^21 mu, 14x14 bilinear form ranks; histogram for random V, structured V, 200 random quadratic maps.
- **Falsifier**: No V family differs from null at 5%, or F4 dreg unchanged when low-rank components are added.
- **Lean gap (assumed, not proved)**: Reduction of the Z-identity mod 2 to GF(2^n) (char-2 expansion), the S_3 form (x1x2+x1x3+x2x3)^2+x1x2x3+b for the binary curve, the rank bound <=2d, and the empirical rank spectrum vs random-map null (>=30% structured cases).

H039 — Heavy Walsh coefficient of the decomposability indicator as target prefilter

- **Handbook**: 9.1.2 Walsh transform, 9.3.2 Bent functions · **Prior art**: partially-tried · **Lean**: proved (`H039_claim`; theorems: `H039_list_sum_le`, `H039_floor`, `H039_parseval_instance`, `H039_claim_holds`)
- **Claim**: The decomposability indicator D of a subspace base has a normalised Walsh coefficient several times the random null near V-perp-related u; targets selected by that parity decompose more often. Trace and learned prefilters were null; this tests the full spectrum.
- **Prediction**: Max coefficient >=3x null on >=70% of curves; favoured-class success >=1.2x; else <=1.5x (null).
- **Cheapest test**: n=20: exact D on all targets, FWHT; compare to 200 random equal-density sets and the trace-parity positive control.
- **Falsifier**: Max within 1.5x null on >=70% of curves, or favoured-class gain <1.1x.
- **Lean gap (assumed, not proved)**: Parseval sum_u W(u)^2 = N|D| from character orthogonality on F_2^n (taken as hypothesis hP; only verified by decide on one n=3 instance), and the actual decomposability indicator D of a curve's subspace base.

H040 — Low-degree annihilators of the decomposability indicator

- **Handbook**: 9.1.6.3 Optimal algebraic immunity · **Prior art**: partially-tried · **Lean**: proved (`H040_claim`; theorems: `H040_mon_succ_le`, `H040_mon_mono`, `H040_instance14`, `H040_instance16`, `H040_annihilator_of_count`, `H040_count_persists`, `H040_claim_holds`)
- **Claim**: For subspace or invariant V at toy scale, algebraic immunity of D is >=2 below ceil(n/2), and the annihilator test rejects a constant fraction of undecomposable targets; degree e may track F4 dreg.
- **Prediction**: AI(D)<=ceil(n/2)-2 for >=80% of (curve,V); rejects >=20% of undecomposable targets; random sets AI >= ceil(n/2)-1.

- **Cheapest test**: Exhaustive D for n=14..18; find AI by F2 linear algebra on monomials of degree <=e; 100 random equal-density sets; compare e with F4 dreg.
- **Falsifier**: AI within 1 of ceil(n/2) on >=80% of cases, or rejection <5%.
- **Lean gap (assumed, not proved)**: The GF(2) linear-algebra step that more degree-<=e monomials than |supp D| yields a nonzero annihilator (hypothesis hlin), the identity sum_{i<=e}C(n,i)=2^n at e=n, the actual |supp D| of curve indicators, and the rejection-rate/F4 d_reg link.

H041 — Trace-form isotropic block pairs giving free linear equations

- **Handbook**: 7.2.3 Trace forms · **Prior art**: partially-tried · **Lean**: proved (`H041_claim`; theorems: `H041_reduction_factor`, `H041_strict`, `H041_claim_holds`, `H041_mono`)
- **Claim**: S3 coordinates are Tr(mu s t) bilinear forms; choosing V1 in (mu V2)^perp for a subspace M of mu kills the bilinear part on dim M coordinates, so first-fall degree drops >=2 and F4 time >=4x vs random pairs. MinRank family is open.
- **Prediction**: n=26,d1=d2=13: random dim M about 0; isotropic pairs 4-8; F4 ratio about 5x (2x-30x).
- **Cheapest test**: n=26: random and isotropic V1,V2; rank of 26 Gram matrices; msolve/F4 on 30 x_R each with matched solution counts.
- **Falsifier**: dim M no larger than random, or F4 ratio <1.5x.
- **Lean gap (assumed, not proved)**: That the isotropic choice V1 in (mu V2)^perp actually yields k = dim M independent affine equations (Gram-matrix rank over F_2 for the trace forms) and that F4 time scales with the enumeration bound. The Lean file proves only the 2^u -> 2^(u-k) arithmetic under k <= u.

H042 — Exact character-sum yield ranking of cyclic-code subspaces

- **Handbook**: 6.2.2 Additive character sums, 6.2.4 Generic estimates, 6.3.3.3 Dual BCH distance · **Prior art**: partially-tried · **Lean**: proved (`H042_claim`; theorems: `H042_orthogonality_4`, `H042_indicator_expansion_4`, `H042_ratio_needs_eps`)
- **Claim**: Frobenius-invariant subspaces of equal dimension m about n/3 on one Koblitz curve have yield N(V) differing by >=3 sigma_binomial; top decile is >=1.2x median and the ranking transfers between a=0 and a=1 curves. Coset indicators were random-level.
- **Prediction**: n=35: best/median 1.15-1.6; Spearman across curves >0.5.
- **Cheapest test**: n=35 (or 21,45): all divisors g with dim 10-13; exact N(V) on 3 Koblitz toys; ranked table with binomial intervals; A/A split.
- **Falsifier**: All N(V)/E within 2 sigma, or cross-curve Spearman <0.2.
- **Lean gap (assumed, not proved)**: The Weil/binomial sigma for the cyclic-code subspace yields on Koblitz curves, the exact N(V) as a character sum over the curve, and the cross-curve (a=0 vs a=1) rank transfer. Orthogonality and indicator expansion are checked only by decide on 4-bit strings.

H043 — Sparse Gauss-period normal basis for the restricted Semaev system

- **Handbook**: 5.3.2 Gauss periods, 5.3.1 Optimal normal bases · **Prior art**: tried-open · **Lean**: proved (`H043_claim`; theorems: `H043_complexity_d5`, `H043_complexity_d7`, `H043_terms_le_two`, `H043_cx_le`)

- **Claim**: In a type-II ONB the multiplication tensor has about 2d-1 nonzeros, so restricted S3 has <=0.5x monomials and F4 time falls, with dreg unchanged; G3 sparse-ONB hypothesis is open and unmeasured.
- **Prediction**: Monomial ratio 0.3-0.6 at d=5..7,q=2^5; F4 time ratio 0.4-0.9.
- **Cheapest test**: d=5,q=2^5 (n=25): 3d-variable S3 in polynomial vs ONB basis, same F4 and seeds, interleaved, 20 repeats.
- **Falsifier**: Monomial ratio >0.8 or F4 time ratio >1.0 at d=5 and d=7.
- **Lean gap (assumed, not proved)**: That the abstract fold model (b_1*b_j = b_{j+1}+b_{j-1}, zero index drops) is the real type-II ONB multiplication tensor for F_{2^d} with 2d+1 prime and 2 primitive, the general equality complexity = 2d-1 (only decided for d=5,7; the general upper bound 2d is proved), and the monomial and F4 time ratios of the restricted Semaev system.

relation-collection (12)

H044 — Order-6 automorphism-orbit quotient for j=0 curves, q=1 mod 3

- **Handbook**: 16.6.5 Endomorphisms, 16.6.4 Restriction of scalars · **Prior art**: tried-open · **Lean**: proved (`H044_claim`; theorems: `H044_twisted_consistent`, `H044_order6`, `H044_ratio`, `H044_claim_holds`, `H044_total_improves`)
- **Claim**: For $y^{2=x}3+b$, S_m is weighted-homogeneous so F_q-subspace bases are closed under x->wx and negation; quotienting by order 6 needs B/6 relations vs B/2, cutting relations and LA size 3x. Matches unrun DGM-01 ablation.
- **Prediction**: q=97,n=3: relations ratio (negation-only)/(order-6) = 3.0+-10%; dreg not increased.
- **Cheapest test**: Existing Diem-style IC on a j=0 toy twice (negation-only vs order-6 reps), same V; compare relations to full rank.
- **Falsifier**: Ratio <2.5, or lambda-twisted signs make logs inconsistent mod l.
- **Lean gap (assumed, not proved)**: That for $y^{2=x}3+b$ with q = 1 mod 3 the Semaev summation basis is closed under x -> wx and negation, the automorphism of order 6 acts on logs by multiplication by lam with lam^6 = 1 mod n (taken as a hypothesis), and the quotiented relations remain independent (full rank) with dreg not increased.

H045 — Stop collection at orientability threshold; target-only left-kernel extraction

- **Handbook**: 15.3 LDPC and Gallager codes, 15.5.1 Tornado codes, 17.3.8 Coding over networks, 15.5.2 LT and fountain codes · **Prior art**: partially-tried · **Lean**: proved (`H045_claim`; theorems: `H045_extract`, `H045_claim_holds`, `H045_determined_mono`)
- **Claim**: t=log_P Q is determined at the first left-kernel vector y of the factor-base part with y.b nonzero, at N/B about 0.918 (m=3), 0.5 (m=2), below B+1; online echelon stopping at target-determined (N_det vs N_full) is the same rule. Solver contract exists, N_det unmeasured.
- **Prediction**: N_dep/B = 0.92+-0.03 (m=3), about 0.52 (m=2); about 8% saving for m=3, about 2x for m=2.
- **Cheapest test**: Koblitz n=23..29, m=3, Frobenius-orbit columns: prefix ranks mod ell on real relations; first N with rank<N and y.b!=0; check recovered t; 20 seeds.
- **Falsifier**: Median N_dep/B outside [0.88,0.96], wrong t, or left-kernel solve costs more than relations saved.

- **Lean gap (assumed, not proved)**: The hypothesis (Ya + Yb*x) = 0 mod n, i.e. summing the relations along a left-kernel vector (swap of the double sum over rows and columns of the factor-base logs), that real Koblitz orbit relations reach a dependency at N/B about 0.92 (m=3), and that left-kernel solving costs less than the relations saved.
- Merged from pool keys: P1-5, P3-7

H046 — Cayley-graph walk candidate generation with lambda_2-selected generators

- **Handbook**: 14.10.3 Expander graphs, 14.10.4 Cayley graphs · **Prior art**: partially-tried · **Lean**: proved (`H046_claim`; theorems: `H046_walk_sound`, `H046_claim_holds`, `H046_ops_per_relation`)
- **Claim**: Walking R_{i+1}=R_i+s_j on Cay(Z_N,S) with r=8-16 generators yields relations per candidate equal to iid (a,b) within 3% at >=5x fewer group-op equivalents per relation for m=2; no gain if the baseline already walks.
- **Prediction**: Yield ratio 0.97-1.03; lag-1 autocorrelation <0.01; duplicates <0.1% over 10^7 steps; ops 5-9x fewer.
- **Cheapest test**: Prime-field curve of order about 2^24: iid control, walks r=4,8,16 with random S, r=16 with lambda_2-minimised S; count relations, rank at R=B+20, ops.
- **Falsifier**: Yield ratio <0.9, rank deficient in >1% of runs, or autocorrelation >0.05 at r>=8.
- **Lean gap (assumed, not proved)**: The expander mixing property (yield ratio within 3% of iid, lag-1 autocorrelation, duplicate rate) from lambda_2 of Cay(Z_N,S), encoded only as the hypothesis h046Mixing, and the group-op cost model (iid candidate at least 6 ops versus 1 mixed addition).

H047 — Koblitz tau-adic Gray-code enumeration with a_0=1 orbit-duplicate filter

- **Handbook**: 14.8.5 Digital (t,s)-sequences, 14.8.6 Gray-code enumeration · **Prior art**: partially-tried · **Lean**: proved (`H047_claim`; theorems: `H047_gray_injective_6`, `H047_gray_onebit_8`, `H047_gray_closed_8`, `H047_filter_half_6`, `H047_gray_cost`)
- **Claim**: Gray-code enumeration of tau-adic strings with a_0=1 costs one mixed addition per candidate, orbit-duplicate rate 0 (50% unfiltered), and yield equal to iid within 3%.
- **Prediction**: Duplicates 50+-1% unfiltered at k=20, 0% filtered; yield ratio 0.97-1.03; about 1 addition vs 2n/3 per candidate.
- **Cheapest test**: Koblitz a=1, n=17,19: enumerate 2^20 strings both ways; count distinct orbits and m=2 relations over a stable base; compare iid tau-adic digits.
- **Falsifier**: Duplicates >1% after filter, yield ratio outside [0.9,1.1], or rank growth lower by >1%.
- **Lean gap (assumed, not proved)**: The general-k statement (gray injective, one-bit steps, closed on 2^k, half of strings with a_0=1) is only decided for k=6 or 8; the identification of the tau-adic orbit-duplicate structure (a_0=1 removes Frobenius duplicates) and the incremental point update equal to the closed form tau-adic expansion on a Koblitz curve; yield equal to iid within 3%.

H048 — Walsh-bias audit of the decomposable-target indicator for linear prefilters

- **Handbook**: 14.2.3 Directions and Redei polynomials, 14.2.4 Sets determining few directions · **Prior art**: partially-tried · **Lean**: proved (`H048_claim`; theorems: `H048_parseval_3`, `H048_floor`)
- **Claim**: For random V the max normalised Walsh coefficient of D_V stays within random-set fluctuation; bias beyond that exists only for inversion- or Frobenius-closed V (e.g. alpha=1). Trace and weight prefilters were null.
- **Prediction**: Random V max |W|/|D| about 0.01-0.02 at n=22; inversion-closed V 0.02-0.1; yield gain <=1.02-1.1x, likely null.
- **Cheapest test**: n=20-22: FFT of D_V for 20 random, 20 invariant, 20 inversion-closed V vs random sets of equal density; planted-correlation positive control.
- **Falsifier**: No coefficient above the 4-sigma null band for any family, or detector fails the planted control.
- **Lean gap (assumed, not proved)**: That the decomposable-target indicator D_V of a curve (Semaev summation polynomial solvability on V) is a set to which Parseval applies, the random-set Walsh fluctuation band, and the existence of bias for inversion- or Frobenius-closed V. Parseval is decided only for all subsets of the 3-bit cube; the floor lemma is proved for arbitrary lists.

H049 — Streaming PLE rank certificate to stop at rank B-1 versus fixed excess

- **Handbook**: 13.4.2.2 PLE decomposition, 13.4.4.2 Rank, determinant, characteristic polynomial · **Prior art**: partially-tried · **Lean**: proved (`H049_claim`; theorems: `H049_stop_le_fixed`, `H049_net_improvement`, `H049_priced_saving`)
- **Claim**: Certifying rank B-1 online stops collection 4-12% below a fixed 1.05-1.1 B excess and always solves; the incremental interface exists but the saving is unmeasured.
- **Prediction**: Median saving about 7%; relations phase falls the same fraction.
- **Cheapest test**: Replay stored toy streams; rank mod ell per 1% increment; first count with rank=B-1 vs 1.1B; 30 seeds per curve, paired; charge probe cost at 0/0.1/1 GAE.
- **Falsifier**: Median saving <2%, or probe cost exceeds saved relation cost.
- **Lean gap (assumed, not proved)**: That rankAt (rank mod ell of the first k collected relations) reaches B-1 below the fixed 1.05-1.1 B excess, i.e. the hypothesis B-1 <= r K at the fixed time K, with a streaming PLE probe whose cost is the separate priced field; the 4-12% saving is empirical.

H050 — Fano factor of solution counts: Frobenius clumping of yield per call

- **Handbook**: 11.2.3.2 Factorization patterns, 11.2.3.5 Permutations and integers · **Prior art**: partially-tried · **Lean**: proved (`H050_claim`; theorems: `H050_solutions_le`, `H050_orbits_le`, `H050_claim_holds`)
- **Claim**: On invariant V, solutions per target are compound-Poisson with Fano factor about n, so independent relations per call are orbits, not solutions; yield accounting must divide by orbit multiplicity.
- **Prediction**: n=23,m=3: Fano 15-23 invariant, 0.9-1.2 non-invariant; mean orbits per call within 20% of non-invariant mean.

- **Cheapest test**: S_3 for 2000 targets with normal-basis-span V vs random V of equal dimension; record solutions, orbits, variance/mean.
- **Falsifier**: Invariant-V Fano below 3, or non-invariant Fano above 2.
- **Lean gap (assumed, not proved)**: That solutions on a Frobenius-invariant V decompose into orbits of size dividing n (orbit sizes taken as the input list with 1 <= s <= k), and the compound-Poisson Fano factor about n and non-invariant Fano about 1, which are not formalised.

H051 — Low-discrepancy lattice sweeps over (a,b) at high saturation

- **Handbook**: 10.4.5.5 Discrepancy, 10.4.5.2 Lattice test · **Prior art**: untried · **Lean**: proved (`H051_claim`; theorems: `H051_dups_zero_of_nodup`, `H051_korobov_nodup`, `H051_korobov_dups_zero`, `H051_dups_instance`)
- **Claim**: A Korobov rank-1 lattice over (a,b) has fewer duplicate relations and faster rank growth than iid when relations needed are >=10% of the effective space; no difference below 1% saturation.
- **Prediction**: At 20% saturation duplicates 10%->under 3%, relations to full rank -5-8%; at 1% difference <0.5%.
- **Cheapest test**: Toy curve of order about 2^16, B=64: iid vs Korobov vs Sobol, interleaved seeds, A/A, 100 repetitions, paired bootstrap.
- **Falsifier**: Relations-to-full-rank intervals overlap at 20% saturation, or lattice arm has lower rank.
- **Lean gap (assumed, not proved)**: Relation-rank growth and the 10%->3% duplicate rate at 20% saturation are empirical (h051Prediction); the proofs assume the decomposition is a deterministic function of (a,b), so distinct (a,b) give distinct relations. The second conjunct of H051_claim is not provable from the model.

H052 — Multiplier-closed table via commuting x-only maps (Dickson analogue)

- **Handbook**: 9.6.1 Dickson polynomials basics, 9.6.2.2 Maps Phi_a, Psi_a · **Prior art**: partially-tried · **Lean**: proved (`H052_claim`; theorems: `H052_relation_scale`, `H052_four_le_sq`, `H052_cost_lower_bound`, `H052_claim_holds`)
- **Claim**: A table {x(kF_j): k in K} built by composing R_2,R_3 raises relations per tuple by about |K| without enlarging the matrix; cost follows K*B+N/K*, *optimum 2*sqrt(NB) (BSGS-IC hybrid), still above rho.
- **Prediction**: GF($2^{20)}$,B=2 7,|K|=16: relations per 10^6 tuples x16+-3; rank >= B-2; cost 2^14.5 vs 2^20.
- **Cheapest test**: GF(2^20), m=3, B=128: K=1,4,16,64; relations per tuple, table cost, rank; compare against a plain base of size K*B at equal budget.
- **Falsifier**: Yield gain <|K|/2, rank deficiency >3, or total cost at optimal K not >=4x below K=1.
- **Lean gap (assumed, not proved)**: That the table entries x(kF_j) have logs k*log F_j (commuting Dickson R_2/R_3 maps acting as scalar multiplication on the x-line), and the relations-per-tuple gain of about |K| and rank >= B-2. Only the exponent-side coefficient sum and the (*K*B+M)^2 >= 4NB bound are proved.

H053 — Consecutive-scalar sampling with pseudorandomness (C_2) guard

- **Handbook**: 6.3.5.2 Measures of pseudorandomness, 6.3.5 Incomplete character sums · **Prior art**: untried · **Lean**: proved (`H053_claim`; theorems:

`H053_seq_le_indep`, `H053_seq_lt_indep`, `H053_speedup`, `H053_claim_holds`)

- **Claim**: Scanning k,k+1,… with R_{k+1}=R_k+P gives hit rate [x(kP) in V] within 3% of independent scalars with no excess correlation, at one addition per candidate instead of a scalar multiplication.
- **Prediction**: Hit-rate ratio 1.00+-0.03; lag-1 correlation within 4/sqrt(hits); candidate cost drop about 20x, end-to-end 5-10x.
- **Cheapest test**: 2^24 consecutive points vs 2^24 independent; hit counts, autocorrelation to lag 16, Mauduit-Sarkozy C_2 for windows $2^{6..2}12$.
- **Falsifier**: Hit-rate ratio outside 0.95-1.05, lag z-score >4, or pair-collision rate differing >5%.
- **Lean gap (assumed, not proved)**: Equal hit rate of consecutive scalars k, k+1, … versus independent scalars (Mauduit-Sarkozy C_2 pseudorandomness, lag correlation) is assumed via the shared expectedTrials count. The addition-cost versus scalar-multiplication-cost gap is a hypothesis (add < mul).

H054 — Cubic/sextic Jacobi-sum j=0 panels with mu_3/mu_6 factor-base quotient

- **Handbook**: 6.1.2 Jacobi and Gauss sums of small orders · **Prior art**: partially-tried · **Lean**: proved (`H054_claim`; theorems: `H054_cols_exact`, `H054_cube_action`, `H054_rewrite`, `H054_la_ninth`, `H054_claim_holds`)
- **Claim**: On j=0 toys, quotienting by mu_3 on top of negation (log lifted by lambda^2+lambda+1=0) cuts relations, search cost and LA size 3x; Jacobi sums choose prime-order test curves. Overlaps order-6 idea; this adds lambda-lifted logs.
- **Prediction**: Relation time to full rank falls 2.5-3.2x vs negation-only at p $2^{28..2}36$; rank deficiency <2; logs match baseline 100/100.
- **Cheapest test**: p about 2^30, p=1 mod 3, prime-order b via Jacobi sum: baseline orbit key vs mu_3 key with lambda in the matrix; 10 seeds interleaved; validate vs rho.
- **Falsifier**: Relations fall <2x, any wrong log, or baseline already uses mu_3.
- **Lean gap (assumed, not proved)**: That the j=0 curve has a mu_3 automorphism acting on points with log(act P) = lam*log P and lam^3 = 1 mod n (and the primality and order facts from the Jacobi sum). The orbit action being free on the factor base and the 2.5-3.2x relation-time gain are assumed or empirical.

H055 — Gray-code incremental products over a subspace factor base

- **Handbook**: 5.3.5 Fast arithmetic using normal bases · **Prior art**: partially-tried · **Lean**: proved (`H055_claim`; theorems: `H055_incremental_eq`, `H055_mult_saving`, `H055_gray_visits_once_m4`, `H055_gray_single_flip_m4`, `H055_claim_holds`)
- **Claim**: Enumerating x1 in V in Gray order updates p=x1*x2 and s=x1+x2 by XOR, cutting inner-loop multiplications from about 5 to 4 per pair and time 10-20% at m>=16; same relation list.
- **Prediction**: Counted mults 4.0 vs 5.0; throughput 1.1-1.25x independent of n in [53,131].
- **Cheapest test**: n=53,m=16: pair loop with/without Gray enumeration; instrumented field type; 20 interleaved repeats; identical relation lists.
- **Falsifier**: Mult reduction <10%, differing relation lists, or wall-clock gain <1.05x.
- **Lean gap (assumed, not proved)**: That the field multiplication by a fixed element (and the Frobenius or trace maps used) is an XOR-linear map f on bit vectors in the

chosen basis, and the real 5 versus 4 multiplication count and wall-clock gain. Gray bijectivity is only checked by decide for m=4, not proved for all m.

relation-filtering (8)

H056 — Overlap-bounded degree-balanced relation selection before filtering

- **Handbook**: 17.3.5 Sequences with small Hamming correlation · **Prior art**: partially-tried · **Lean**: proved (`H056_claim`; theorems: `H056_finSum_zero`, `H056_finSum_single`, `H056_dot_unit`, `H056_zero_coverage_kernel`, `H056_RS_design_q5`, `H056_RS_overlap_q3`, `H056_claim_holds_fragment`)
- **Claim**: At fixed B, RS-profile greedy selection (balanced column coverage, capped pairwise overlap) needs >=5% fewer relations for full column rank than uniform subsampling of the same pool, with no heavier matrix.
- **Prediction**: 3-12% fewer (central 6%); gain shrinks with pool size; zero when pool equals needed count.
- **Cheapest test**: n=23, subspace FB dim 11 (B about 2000), pool 6000 weight-3 relations, 200 seeds: smallest N for rank B-1 mod 31-bit prime, uniform vs greedy; A/A shuffle.
- **Falsifier**: Paired mean reduction <2%, or CI includes 0 while A/A centres at 0.
- **Lean gap (assumed, not proved)**: Relations are modelled as integer row vectors; rank over a 31-bit prime field and the real RS-profile greedy selector, the 5-6% reduction in relations needed, and the gain shrinking with pool size are empirical. RS design properties are checked by decide for q=5 (coverage) and q=3 (overlap) only. H056_claim's second conjunct is only the weak fragment N_greedy <= N_uniform.

H057 — Density-evolution threshold for filtering overhead (non-uniform degrees)

- **Handbook**: 16.3.3.12 Sparse linear solving, coding-theory cross-over · **Prior art**: partially-tried · **Lean**: proved (`H057_claim`; theorems: `H057_cost_transfer`, `H057_claim_holds`, `H057_peel_sound`, `b06_peel_solvable`)
- **Claim**: Core size after peeling matches the density-evolution fixed point within 5% using measured column degrees (2-core at c*=0.818 for m=3); the cost-optimal overhead c is predictable without running LA and differs from the current default.
- **Prediction**: m=3 uniform: core fraction 0.45-0.60 at c=1.02; within 3% at c=1.2; measured-degree prediction within 5%.
- **Cheapest test**: Archived toy sets n=23..29: sweep c 0.8-1.3 by subsampling, measure core after peeling, compare with recursion using measured degrees.
- **Falsifier**: Core deviates >15% from prediction, or filtering already within 2% of optimal c.
- **Lean gap (assumed, not proved)**: The density-evolution recursion and its fixed point (c* = 0.818 for m=3) and the match of measured core size within 5% are not formalised; they enter as the hypothesis h057Within 5 core pred. Only the transfer of core-size error to dense-phase cost, peel soundness and the termination measure are proved.

H058 — Closed-form 2-core as a hidden-structure diagnostic

- **Handbook**: 15.5.2 LT and fountain codes, 15.5.3 Raptor codes · **Prior art**: partially-tried · **Lean**: proved (`H058_claim`; theorems: `H058_core_le`, `H058_claim_holds`, `H058_classify_sound`, `H058_classify_instance`, `H058_peel_preserves`)
- **Claim**: On real m=3 matrices post-peeling core fraction equals density evolution (0.72 at c=1.0, 0.85 at c=1.2) within 3 points; deficits measure engineerable structure (orbits, symmetrisation), excess measures hidden bias.
- **Prediction**: Core 0.72+-0.03 at c=1.0, 0.85+-0.03 at c=1.2; orbit-merged columns >=5 points smaller.
- **Cheapest test**: n=31 Koblitz, m=3, B about 2000: run singleton filter and compare with formula; 20 seeds.
- **Falsifier**: Core within +-1 point (nothing to engineer), or >5 points off both ways without cause.
- **Lean gap (assumed, not proved)**: The closed-form core fractions (0.72 at c=1.0, 0.85 at c=1.2) and the interpretation of deficits as orbit or symmetrisation structure and excess as hidden bias are empirical. The bound 100*core* <= *f*B is a hypothesis. Peeling is modelled for unit-coefficient singleton columns over Int mod m only.

H059 — Closed-form peeling core with max-degree inactivation ordering

- **Handbook**: 15.5 Raptor codes, 15.5.1 Tornado codes, 15.3 LDPC density evolution · **Prior art**: partially-tried · **Lean**: proved (`H059_claim`; theorems: `H059_wiedemann_gain`, `H059_claim_holds`, `H059_peel_backsub`, `H059_inactivate_max`, `b06_backsub_unique`)
- **Claim**: For m=3 the core is 0.635B at N=0.918B and 0.723B at N=B; peel plus Raptor inactivation then Wiedemann on the core costs >=1.5x fewer modmuls than on the unfiltered matrix. Risk: relabels structured elimination.
- **Prediction**: Core within 3% (random), 8% (real); core Wiedemann 0.4-0.7 of unfiltered count.
- **Cheapest test**: Peeler on saved toy relations (n about 27, B $2^{12..2}14$): core after peeling and inactivation; block Wiedemann modmuls vs structured GE at equal weight.
- **Falsifier**: Core off by >8% on real data, or core-only modmuls not below 0.9 of structured-GE + Wiedemann.
- **Lean gap (assumed, not proved)**: The core fractions 0.635B at N=0.918B and 0.723B at N=B (here 1000*core* <= *723*B is a hypothesis), the Wiedemann cost model d*d*w versus the real block Wiedemann modmul count, and the claim that inactivation does not just relabel structured elimination.

H060 — Inactivation-decoding order replacing greedy singleton/clique removal

- **Handbook**: 14.4.6 k-Arcs and linear MDS codes · **Prior art**: partially-tried · **Lean**: proved (`H060_claim`; theorems: `H060_dense_cubic`, `H060_inactivation_perm`, `H060_claim_holds`, `H060_unit_peel`, `b06_perm_solvable`)
- **Claim**: Inactivating the max-residual-weight column when peeling stalls leaves a dense core >=10% smaller than singleton-plus-clique filtering at oversampling 1.1, lowering Lanczos cost; Markowitz pivoting is the nearest prior art.
- **Prediction**: Core -15-40% at row weight 4-6; dense-phase cost -30-60%; whole-solve effect small.

- **Cheapest test**: Toy matrices n=20-24 orbit-quotiented: singleton-clique vs max-weight inactivation; compare dense core dimension, total ops, rank mod l.
- **Falsifier**: Core not >=10% smaller, or rank differs after filtering.
- **Lean gap (assumed, not proved)**: That max-residual-weight inactivation yields a core at least 10% smaller than singleton-plus-clique filtering (10*dIn* <= *9*dSC is a hypothesis) at oversampling 1.1, and the dense-phase cost being cubic in the core dimension. Only the cost monotonicity, solution-set invariance under column permutation and unit-coefficient peel soundness are proved.

H061 — Z/4 torsion character from halving to prune sign patterns and reject corrupt rows

- **Handbook**: 12.2.6 Torsion and Tate module, 12.2.7 Weil and Tate pairing · **Prior art**: partially-tried · **Lean**: proved (`H061_claim`; theorems: `H061_relation_character`, `H061_patterns_length`, `H061_survivors_m3`, `H061_survivors_m4`, `H061_survivors_half_m3`)
- **Claim**: On cofactor-4 curves psi4: E(F_q)->Z/4 (two halvings) is a homomorphism; checking sum eps_i psi4(P_i)=psi4(R) mod 4 prunes about half the sign patterns and rejects corrupt or twist-contaminated rows with probability 3/4 before group ops or LA.
- **Prediction**: Survivors 0.35-0.6 of 2^m; sign-resolution additions down 1.5-2x; corrupt rows rejected about 0.75.
- **Cheapest test**: a=0 Koblitz n=31: homomorphism check on random pairs; 10^5 brute-force m=3 relations, true vs corrupted; a=1 (h=2) control.
- **Falsifier**: Homomorphism fails, survivors >0.6, corrupt rejection <60%, or psi4 cost exceeds saving.
- **Lean gap (assumed, not proved)**: Existence of the Z/4 homomorphism psi4 via two halvings on cofactor-4 curves (modelled as h061Hom). The general 2^(m-1) survivor count for odd psi4 is stated in the claim but only proved for m=3 and m=4 instances by decide.

H062 — Coupon-collector relation count and orbit-quotient coverage law, with rare-column steering

- **Handbook**: 11.6.10 Sparse linear systems, 11.2.3.5 Permutations and integers · **Prior art**: partially-tried · **Lean**: proved (`H062_claim`; theorems: `H062_flatten_len`, `H062_cover_bound`, `H062_orbit_cover_bound`, `H062_claim_holds`)
- **Claim**: Relations to full rank follow (B/m)(ln B+c), and after orbit quotient (B_orb/m)ln B_orb, so saving is n*lnB/ln(B/n), not n (subfield arm measured closed). Fixing x1 to a rare column may lower total solver calls; the Koblitz arm and steering are open.
- **Prediction**: n=24-30,m=3: 3-5B relations to full rank; steering 1.3-2x fewer solver calls or cancelled; quotient saving within 15% of n*lnB/ln(B/n).
- **Cheapest test**: 3-5 curves, n in {22,24,26}: log column sets in arrival order, incremental rank mod group order, fit B/m(lnB+c); folded matrix; steered arm, 20 seeds, count solver calls.
- **Falsifier**: Full rank at <=1.15B unsteered, steering needs >= unsteered solver calls, or quotient saving equals n within 5%.
- **Lean gap (assumed, not proved)**: The coupon-collector (B/m)(ln B + c) law and the steering gain are empirical. Not modelled: that relations are random m-column supports, that a surjective fold from the curve's automorphism or Frobenius action gives B/k orbit columns, and that rank mod the group order equals support coverage.

- Merged from pool keys: P9-1

H063 — Run-structure canonicalisation of normal-basis necklaces in filtering

- **Handbook**: 10.2.4 Run property of LFSR sequences, 10.3.2 Autocorrelation of sequences · **Prior art**: partially-tried · **Lean**: proved (`H063_claim`; theorems: `H063_rot_length`, `H063_lexLe_lz`, `H063_claim_holds`, `H063_instance`)
- **Claim**: The minimal rotation starts at a maximal zero run, so clz/ctz plus 1-2 comparisons canonicalise 131-bit necklaces >=10x faster than naive and 3x faster than Booth, identical outputs. A related arm gave an unresolved 8% instruction saving.
- **Prediction**: Mean candidates 1.6+-0.4; 15-40x over naive, 3-8x over Booth.
- **Cheapest test**: C microbenchmark, 10^7 random 131-bit vectors: naive, Booth, run-based; assert identical outputs; cycles and candidate histogram; real toy x-coordinates.
- **Falsifier**: Any output differing from naive, mean candidates >4, or speedup over Booth <1.5x.
- **Lean gap (assumed, not proved)**: That normal-basis x-coordinates are 131-bit necklaces under Frobenius rotation. The proved result is that a lexicographic minimum has a maximal leading zero run. The candidate-count (about 1.6) and speedup over Booth are not proved.

linear-algebra (4)

H064 — Sparse fountain-coded SpMV for block Wiedemann on Spot

- **Handbook**: 17.3.8 Coding over networks · **Prior art**: partially-tried · **Lean**: proved (`H064_claim`; theorems: `H064_cost_lt_replication`, `H064_decode_any_two`, `H064_claim_holds`)
- **Claim**: Sparse-coded SpMV with redundancy 1.25 keeps expected iteration time <=1.15x no-failure and at preemption q=0.1 beats 2x replication and checkpoint-restart in core-seconds; orchestration saving, not a mathematical speedup.
- **Prediction**: Flops 1.3-1.6x; break-even q about 0.03-0.08; gain <=1.4x at q=0.1, none at q<0.03.
- **Cheapest test**: Simulate B=20000,w=6,k=16 mod 61-bit prime, n=20 workers, q in {0.01,0.03,0.1,0.3}; compare restart, replication, dense RS, sparse fountain; verify decode.
- **Falsifier**: Coded cost >= min(replication,restart) at q=0.1, or decode failure >1e-3.
- **Lean gap (assumed, not proved)**: Full column rank of any received selection of sparse fountain-coded rows (hdec), the decode-failure probability below 1e-3, the preemption model q, and the restart and checkpoint costs. Only the flop-factor versus replication arithmetic and a 2+1 parity decode are proved.

H065 — Block Wiedemann early stop by confirmed linear-complexity profile

- **Handbook**: 17.2.4 Period finding, 17.3.1 Sequences · **Prior art**: partially-tried · **Lean**: proved (`H065_claim`; theorems: `H065_first_ge`, `H065_first_le`,

`H065_first_mono`, `H065_linComp_mono`, `H065_linComp_le`, `H065_saving`, `H065_claim_holds`)

- **Claim**: Online Berlekamp-Massey with a stability window delta stops at about 2L+*delta; on filtered IC matrices L* is >=3% below N/m+N/n, saving >=3% of SpMVs with failure probability <1e-6.
- **Prediction**: Savings 0-4% (central 1%); null most likely.
- **Cheapest test**: B=3000 weight-3 matrix mod 31-bit prime after clique filtering; scalar and block (m=n=4) Wiedemann with online BM; compare iterations over 100 matrices; verify kernel vector.
- **Falsifier**: Median saving <2%, or an early-stopped run returns a wrong kernel vector.
- **Lean gap (assumed, not proved)**: That ok t L (a recurrence of order L explains the first t terms) is prefix-closed for the actual Wiedemann or Berlekamp-Massey sequence, that the minimal-polynomial degree is at most N, that the stability window delta gives failure below 1e-6, and that L* is at least 3% below N/m+N/n on filtered IC matrices (empirical).

H066 — Functional-only Wiedemann reading kernel coordinates from the BM sequence

- **Handbook**: 16.2.1 LFSR synthesis, linear complexity, Berlekamp-Massey · **Prior art**: partially-tried · **Lean**: proved (`H066_claim`; theorems: `H066_functional_comb`, `H066_matvec_count`, `H066_claim_holds`)
- **Claim**: Recovering log(Q) via coordinate functionals u=e_i and a normaliser uses 2N+O(1) matvecs instead of 3N, a 33% cut with identical output.
- **Prediction**: Matvec ratio 0.67+-0.02; log(Q) identical in 100% of 200 runs.
- **Cheapest test**: 3000x3000 weight-3 matrix mod 40-bit prime from an n=31 curve: standard Wiedemann kernel vs functional variant; matvecs and log(Q).
- **Falsifier**: Scalar BM fails to give the minimal polynomial in >5% of runs, or ratio >0.8.
- **Lean gap (assumed, not proved)**: That the BM minimal polynomial of the scalar sequence u(A^j b) is the true annihilator, that the coordinate functionals u=e_i with a normaliser recover log(Q) for the actual relation matrix mod the group order, and the 2N+O(1) count of the O(1) overhead. Only linearity of u over a combination and the matvec arithmetic are proved.

H067 — Displacement-rank solve of the un-quotiented Frobenius-twisted relation matrix

- **Handbook**: 13.4.5.2 Structured matrices and displacement rank, 13.2.6 Hankel and Toeplitz · **Prior art**: partially-tried · **Lean**: proved (`H067_claim`; theorems: `H067_count_le`, `H067_claim_holds`, `H067_instance`)
- **Claim**: After full orbit expansion, M-ZMD has constant generator rank (<=8) independent of B, enabling O(alpha^2 B) Toeplitz-like solves; relation supports chosen by smoothness may carry no shift structure.
- **Prediction**: Median displacement rank <=8 for n in {17,19,23}, B $2^{8..2}11$; random control >=0.45B; solve cost down >=10x at B=2^11.
- **Cheapest test**: Un-quotient toy Koblitz runs, reduce mod ell, compute rank(M-ZMD) over F_ell; shuffled-support control; seconds for B<=2048.
- **Falsifier**: Rank grows linearly (slope >=0.2) and matches the shuffled control.
- **Lean gap (assumed, not proved)**: That the un-quotiented Frobenius-twisted relation matrix is shift-invariant away from orbit boundaries (the twist-closure hypothesis), that smoothness-selected supports keep that structure, and that rank(M-ZMD) is at

most the number of nonzero displacement rows. Matrix rank over F_ell is not defined here, only the nonzero-row count.

descent (7)

H068 — Waring-type coverage threshold for descent steps

- **Handbook**: 13.1.6 The Waring problem over finite fields · **Prior art**: untried · **Lean**: proved (`H068_claim`; theorems: `H068_tuples_length`, `H068_cover_count`, `H068_necessary`, `H068_claim_holds`)
- **Claim**: Representing a target as a sum of k cubes from {v^3: v in V} is complete once km>=n+2log2(n)+O(1); below it a fraction exp(-2^(km-n)) is uncovered, so descent retries are 1+exp(-mu), not geometric.
- **Prediction**: n=17,19,23,m=n/3+2: uncovered fraction matches exp(-mu) within 20% for km-n in [-2,3] and is 0 at about 5.
- **Cheapest test**: n=17,m=8,k=3: N_3(t) for all t by Walsh-Hadamard convolution; compare to Poisson(mu); smallest km-n with zero uncovered.
- **Falsifier**: Variance >1.5x Poisson, or full coverage needs km-n>10.
- **Lean gap (assumed, not proved)**: That the cube set V has 2^m elements and that sums of cubes in F_{2^n} are indexed by t < 2^n. The Poisson law exp(-2^(km-n)) for the uncovered fraction and the sharp threshold km >= n + 2log2(n) are empirical and not proved. Only the necessary condition km >= n is proved.

H069 — Riemann-Roch counting threshold on descent level size

- **Handbook**: 12.1.2 Divisors and Riemann-Roch theorem · **Prior art**: partially-tried · **Lean**: proved (`H069_claim`; theorems: `H069_subsets_length`, `H069_threshold`, `H069_claim_holds`, `H069_instance`)
- **Claim**: An m-term descent step succeeds with probability 1-exp(-C(|V'|,m-1)/q), sharp at |V'|^(m-1)=(m-1)!q; for m=3 the first level needs about sqrt(2q) (rho scale), so descent is cheap only for m>=4. Yield law measured, sharp threshold not.
- **Prediction**: n=24,m=3: 50% crossing near d=12-13; m=4 near d=8-9.
- **Cheapest test**: n=24 toy: all pairs/triples from V' of dim 8..14 into a hash of x-sums; fraction of 10^4 random targets decomposed vs Poisson curve.
- **Falsifier**: Measured threshold dimension differs from the formula by >=1.5 at m=3 or m=4.
- **Lean gap (assumed, not proved)**: That a descent step is an (m-1)-subset sum from V' hashed into the targets, and the Poisson success law 1-exp(-C(|V'|,m-1)/q) with its sharp threshold. The relation to dimension d is also missing: |V'|=2^d. Only the pigeonhole necessity N <= C(|V'|,m-1) is proved.

H070 — Kloosterman/Weil yield bound for codimension-c descent targets

- **Handbook**: 10.1.4.4 Gauss sums, 10.1.3 DFT · **Prior art**: partially-tried · **Lean**: proved (`H070_claim`; theorems: `H070_indicator_expansion`, `H070_single_bit`, `H070_claim_holds`)
- **Claim**: Descent yield from V1 to V' is $|V1|^{2^{2}\wedge(-c)(1+eps) \text{ with } |eps| <= C^{2}}$c*sqrt(q)/|V1|^2, C<=4; at |V1|=q^(1/3) bias decays like q^(-1/6), so exploitable bias exists only at toy sizes.

- **Prediction**: n=19..23,d=6..8,c=1..3: |eps| below the bound; >10% only when the bound exceeds 10%; median |eps|<3% at d>=n/3.
- **Cheapest test**: Exhaustive counts of (x1,x2) in V1^2 with x3 in V' for 300 random (V1,V',beta) triples at n=19..23.
- **Falsifier**: Any |eps| above the C=4 bound, or persistent bias >10% at d>=n/3 for structured V1.
- **Lean gap (assumed, not proved)**: The Weil/Kloosterman bound on the non-trivial character sums (C <= 4, |eps| <= C*2^csqrt(q)/|V1|^2), identification of the parity bits with trace-form characters of the subspace V', and the pair-sum structure of N over V1^2. Only the product-of-characters indicator expansion of a codimension-c parity condition is proved.

H071 — Bilinear dreg and time of the 2-point descent system

- **Handbook**: 9.1.1.1 Algebraic normal form, 9.2.7 Quadratic APN functions · **Prior art**: partially-tried · **Lean**: proved (`H071_claim`; theorems: `H071_affine_in_x2`, `H071_descent_bound`, `H071_cost_total`, `h071XorSum_xor`)
- **Claim**: With r fixed, S_3 is bilinear in the two bit blocks; F4 dreg <=d+1 is below the generic-quadratic prediction and time per decomposition scales with a smaller base than 2^d enumeration. The bound is published; the measured slope on descent systems is untried.
- **Prediction**: n=17..29, d=ceil(n/2)+2: dreg <=d+1 and >=1 below semi-regular at n>=21; log2 time slope about 0.35n vs 0.5n.
- **Cheapest test**: msolve/F4 on Weil-restricted descent systems for n=17..27 (50 targets); compare with random bilinear, generic quadratic, and enumerate-over-x1 plus F2 algebra.
- **Falsifier**: dreg matches the semi-regular prediction, or time slope per bit >= enumeration at n>=25.
- **Lean gap (assumed, not proved)**: That the Weil-restricted descent system is the bilinear form modelled here, and the measured F4 dreg, semi-regular prediction and log-time slopes (supplied as fields of h071Measure).

H072 — Descent chain chosen by per-level valid-point fraction

- **Handbook**: 9.1.2 Walsh transform, 9.1.3 Parameters of Boolean functions · **Prior art**: partially-tried · **Lean**: proved (`H072_claim`; theorems: `H072_monotone`, `H072_cost_total`)
- **Claim**: Among nested invariant chains V_0<…<V_k of fixed dimensions, per-level success varies beyond binomial noise; the best chain has >=2x fewer expected trials per full descent than the median. A null on the factor-base twin predicts a null here.
- **Prediction**: n=23, dims (6,8,10,12,14), m=2: per-level spread >=1.5x; best-of-500 product of trials <=0.5x median; non-invariant <=0.8.
- **Cheapest test**: 500 random nested invariant chains at n=23; per-level success on 2000 points via exhaustive 2-decomposition; compare with the product of valid fractions.
- **Falsifier**: Per-level spread within binomial noise, or best/median trials ratio >0.8.
- **Lean gap (assumed, not proved)**: Per-level valid-point counts of random nested invariant chains and the median-chain statistic are hypotheses. The Prop assumes the best-of-500 chain exists with at least 2x fewer trials; Lean does not derive per-level spread from curve structure.

H073 — Per-level liftable-bias selection of descent sets

- **Handbook**: 7.1.4 Artin-Schreier hypersurfaces, 6.3.4.2 Inversive congruential pseudorandom numbers · **Prior art**: partially-tried · **Lean**: proved (`H073_claim`; theorems: `H073_biased_le`, `H073_work_mono`)
- **Claim**: Selecting each descent level's V by best-of-T liftable bias cuts attempts per node by (2*liftFrac)^(-m) and total descent cost by the product over levels, with no linear algebra. Shrinks as 2^(-d/2); expected to fail if the factor-base twin fails.
- **Prediction**: n=29, levels d=(8,11,14), T=2^8: liftFrac 0.58, 0.55, 0.53; total cost ratio about 2.5x (1.5-3.5x).
- **Cheapest test**: n=29: 200 random targets, chosen vs random level sets, m=2 closed-form pair enumeration; mean attempts per level, paired bootstrap.
- **Falsifier**: Total attempts ratio <1.15, or per-level ratio not within 20% of (2*liftFrac_measured)^m.
- **Lean gap (assumed, not proved)**: That best-of-T selection raises the liftable fraction above 1/2 and that attempts per node scale as (2 liftFrac)^(-m). Per-level liftFrac, attempts and nodes are record fields; the 1.15x total ratio is asserted.

H074 — Completely normal element for conversion-free subfield-tower descent

- **Handbook**: 5.4.1 Complete normal basis theorem, 5.4.3 Cyclotomic modules and complete generators · **Prior art**: untried · **Lean**: proved (`H074_claim`; theorems: `H074_normal_of_cn`, `H074_nonzero_of_cn`, `H074_six`, `H074_six_zero_not`, `H074_conv_saving`)
- **Claim**: A completely normal beta is normal over every F_{2^d}, d|n, so successive Weil restrictions need zero basis-conversion maps (vs about n^2/64 word ops each) and invariant subspaces nest as kernels ker g_d(sigma). Composite n only; prime n such as 131 excluded.
- **Prediction**: Conversion cost 0; total descent saving <3% at n=36; yield unchanged.
- **Cheapest test**: n=12,36 over F_2: find completely normal beta by checking circulant rank of Tr_{n/d}(beta) at each d | n; count conversions in a three-level descent script vs independent normal elements.
- **Falsifier**: No completely normal element within 1000 trials at n=36 (bug), or descent time differs >5% from baseline.
- **Lean gap (assumed, not proved)**: The model is a GF(2) circulant-rank proxy over the group ring. Real normality of Tr_{n/d}(beta) over F_{2^d} (F_{2^d}-linear independence of conjugates) is not formalised. Existence of a completely normal element at n=36 and the <3% saving are hypotheses. The n=6 instance beta=1 is trivial under the proxy.

cost-model (9)

H075 — Hankel rank spectrum of the trace-component pencil and yield overdispersion

- **Handbook**: 16.3.3.8 Rank attacks, 16.3.3.9 MinRank · **Prior art**: untried · **Lean**: proved (`H075_claim`; theorems: `H075_geom`, `H075_daykin_sum`, `H075_count_zero_two`, `H075_hist_two`, `H075_hist_three`)
- **Claim**: On interval V the component bilinear forms are Hankel with Daykin rank counts 3*2^(n-2d+2r-1); decomposition counts per target are overdispersed vs

Poisson(2^(2d-n)) by the low-rank Walsh spectrum, and low-rank combinations become linear after r branchings.
- **Prediction**: n=19,d=8: rank-1 count 48, rank-2 count 192; Var/Mean>1 by a computable amount at n=17..21.
- **Cheapest test**: n=19,d=8: Hankel rank histogram over all 2^19 delta vs formula; brute-force decomposition counts for all 2^19 targets vs Walsh-sum prediction.
- **Falsifier**: Histogram deviates from 3*2^(n-2d+2r-1) beyond the 2x kernel factor, or Var/Mean equals 1 within error.
- **Lean gap (assumed, not proved)**: Daykin counts for general d are asserted, not proved: the brute-force count equals the formula is proved only by decide at d=2,3. The link from Hankel ranks of the trace-component pencil to per-target decomposition counts, and the Var/Mean>1 overdispersion, are hypotheses.

H076 — Batch-target marginal cost: left-kernel growth about 0.75 per extra relation

- **Handbook**: 15.5.1 Tornado codes, 15.3 LDPC rank above ML threshold · **Prior art**: partially-tried · **Lean**: proved (`H076_claim`; theorems: `H076_batch_le_separate`, `H076_k64`, `H076_adjugate`, `H076_unique`)
- **Claim**: Above the orientability threshold left-kernel dimension grows about 0.74 per extra relation, each dependency giving an equation in the unknown target logs; k targets need about c_3 B+2k relations, so extra targets cost about 2 attempts. Must be priced against matched batch rho.
- **Prediction**: Slope 0.70-0.78 for N/B in [0.92,1.0]; 1.3-2.0 relations per extra target; k=64, B=2000: about 1968 vs 64x1840.
- **Cheapest test**: Random 3-uniform hypergraphs B=2000..8000 mod 61-bit prime: left-kernel dimension vs N, then k=1..64 target-carrying rows; repeat on real Koblitz relations with planted Q_j.
- **Falsifier**: Slope outside [0.55,0.9], >4 relations per target, or real-vs-simulation mismatch >15%.
- **Lean gap (assumed, not proved)**: The kernel-growth slope 0.70-0.78 of random 3-uniform hypergraphs above the orientability threshold, the relation count c3*B, and the bridge from real Koblitz relations to the random model are assumed. Only the k=2 adjugate step is proved, not general k.

H077 — Strength-2 covering array screening of the autolab configuration space

- **Handbook**: 14.7.4 Covering arrays · **Prior art**: untried · **Lean**: proved (`H077_claim`; theorems: `H077_covers`, `H077_main_effects`, `H077_budget`)
- **Claim**: A 6-run binary (or 9-run ternary) strength-2 covering array plus a pairwise model finds a configuration within 10% of the exhaustive best on >=80% of toy curves using <5% of 128 runs; a process-cost tool for finding measured improvements.
- **Prediction**: Within 10% on 80-95% of 50 curves; failures on three-way interactions.
- **Cheapest test**: Full 2^7 grid on 50 toy curves (n=17..23 binary, 20-bit prime) as ground truth; replay the preregistered 6-row array offline with a pairwise model.
- **Falsifier**: <60% of curves within 10%, or median regret >25%.
- **Lean gap (assumed, not proved)**: That the pairwise model fitted on the 6 array rows picks a configuration within 10% of the exhaustive best on at least 80% of curves. This depends on the real cost functions having small three-way interactions; the claim takes pick and best as hypotheses.

H078 — Sidon-defect of Carlitz-torsion and spread-element factor bases

- **Handbook**: 14.5.3 Difference families and BIBDs, 14.4.4 Singer cycles, 13.5.5 Carlitz torsion, 14.3.3 Desarguesian spread · **Prior art**: tried-closed · **Lean**: proved (`H078_claim`; theorems: `H078_pairs_length`, `H078_distinct_le`, `H078_distinct_eq_iff`, `H078_coverage_le`, `H078_coverage_eq_iff`, `H078_tri_two_mul`, `H078_sidon_example`)
- **Claim**: General sumset-coverage and energy were measured null (Sidon base 1.107x only); the untested families are Carlitz/Drinfeld torsion ker(C_a) and spread elements mu*GF(q), predicted to show >3% coverage deficit from structure-induced collisions.
- **Prediction**: Random and spread V within 1% of Poisson; Carlitz-torsion V 0-5% excess collisions, direction uncertain.
- **Cheapest test**: n=20-22, dim-7 subspaces, 50 per family, 5 curves: exhaustive pair-sum x-image size vs Poisson; random-subset negative control; planted AP positive control.
- **Falsifier**: All families within 1% of Poisson, or the positive control shows no defect.
- **Lean gap (assumed, not proved)**: That Carlitz torsion ker(C_a) and spread elements mu*GF(q) subspaces yield x-images with a coverage deviation of at least 1% from Poisson. Mean coverage and expectation are Nat inputs, and the x-coordinate image map is not modelled.

H079 — Delayed-reduction PLE dense core moving the hybrid switch point

- **Handbook**: 13.4.6.1 Hybrid sparse-dense methods, 13.4.1.2 Word size prime fields, 13.4.2.2 PLE decomposition · **Prior art**: partially-tried · **Lean**: proved (`H079_claim`; theorems: `H079_mono_d`, `H079_cost_down`, `H079_argmin_up`)
- **Claim**: With 2-3 limb ell, delayed modular reduction in blocked PLE cuts dense per-entry cost near the block size, moving the optimal dense core >=1.5x larger and total LA cost down >=15%.
- **Prediction**: Core 0.05B->0.09B; LA cost -15-30% at B=$2^{16..2}$20; kernel 3-6x fewer cycles per MAC.
- **Cheapest test**: Microbenchmark 3-limb mul-add reduce-each vs delayed (k=8,16) in blocked PLE of 2048x2048; plug constants into the hybrid cost model, recompute argmin core.
- **Falsifier**: Dense kernel speedup <1.5x, or optimum core moves <1.2x.
- **Lean gap (assumed, not proved)**: The 3-6x delayed-reduction kernel speed-up (3*d' <= 2*d), the sparse cost curve being convex, and the 1.5x core shift and 15% LA cost drop. The strict single-crossing theorem needs a strictly increasing dense cost, while the Prop's 1.5x shift and 15% drop are empirical.

H080 — Hamming-weight degree and mixed-volume predictor for F4 cost

- **Handbook**: 11.5.2.1 Ostrowski's theorem, 11.5.2.2 Polytope indecomposability · **Prior art**: partially-tried · **Lean**: proved (`H080_claim`; theorems: `h080Pop_eq`, `H080_pop_le`, `H080_pop_one`, `H080_pop_add`, `H080_weight_le_total`, `H080_weight_mul`, `H080_hits_le`)
- **Claim**: Bit-degree popcount of exponents plus BKK mixed volume of Weil-restricted supports predicts the F4 maximum degree within +-1 on >=85% of toy instances vs <=65% for the semi-regular bound.

- **Prediction**: Weight predictor 85-95% hit rate; semi-regular 50-65% (n=15..24, m=3, k=n/3, 100 systems).
- **Cheapest test**: 100 Weil-restricted S_3,S_4 systems: run F4, record max degree and matrix size; compute both predictors; compare hit rates (testable from existing degree data).
- **Falsifier**: Weight predictor within 5 points of semi-regular, or hit rate <70%.
- **Lean gap (assumed, not proved)**: The link from the bit-degree weight (and the BKK mixed volume, not modelled) to the F4 maximum degree, plus the hit-rate numbers 85% vs 65%, which come from F4 runs on Weil-restricted systems (hypothesis fromWeilF4Runs).

H081 — Correlation measure C_2 of the decomposability indicator along an additive sweep

- **Handbook**: 10.4.5.3 Correlation measure of order k · **Prior art**: untried · **Lean**: proved (`H081_claim`; theorems: `H081_corr_identity`, `H081_dis_le`, `H081_corr_abs_le`, `H081_C2_abs_le`, `H081_claim_holds`)
- **Claim**: For generic shifts C_2 stays within 3*sqrt(L ln L), but for tau-multiple shifts on Koblitz curves it exceeds it, understating yield variance in the cost model.
- **Prediction**: Generic: sqrt(L) scaling; tau shift excess 2-5x, yield variance 1.5-3x binomial; non-Koblitz no excess.
- **Cheapest test**: Koblitz GF(2^17), B=100: indicator for L=2^16 and order-2 correlation for shifts {1,2,tau-multiples,random}; random-curve null; planted positive control.
- **Falsifier**: C_2 for tau-shifts within the random-curve null band with the positive control detected.
- **Lean gap (assumed, not proved)**: That the decomposability indicator of a Koblitz GF(2^17) additive sweep has the stated generic sqrt(L ln L) band and a tau-shift excess is a measured fact, not proved; the band hypothesis is an explicit premise.

H082 — Multihomogeneous Bezout/mixed-volume predictor of F4 time for split systems

- **Handbook**: 7.1.3 Toric hypersurfaces · **Prior art**: partially-tried · **Lean**: proved (`H082_claim`; theorems: `H082_prod_mono`, `H082_total_sum`, `H082_claim_holds`)
- **Claim**: log F4 time correlates with log B_mh at Spearman >=0.9 vs <=0.6 for total-degree Bezout, so a B_mh-driven model picks the best (d,m,split) more often; counts only if chosen parameters measurably cut time.
- **Prediction**: Spearman 0.9+-0.05 vs 0.55+-0.15; argmin agreement >=70% vs <=40%.
- **Cheapest test**: 200 systems, n in 18..26, d in n/3..n/2, m in {2,3}: msolve/F4 with timeouts; compute both predictors symbolically; Spearman with bootstrap CI.
- **Falsifier**: Spearman gap <0.1 with CI overlapping zero, or argmin agreement no better than Bezout.
- **Lean gap (assumed, not proved)**: Monotonicity of F4 running time in the multihomogeneous Bezout bound (the Spearman >= 0.9 correlation) is a hypothesis; the true mixed-volume bound is replaced by a block-degree product.

H083 — Variance-to-mean (Poisson) audit of decomposition counts via log-domain spectrum

- **Handbook**: 6.3.6.3 Thin sets with small DFT, 6.3.6.4 Character sums over arbitrary sets · **Prior art**: tried-closed · **Lean**: proved (`H083_claim`; theorems: `H083_sums_length`, `H083_coverage_step`, `H083_coverage_mono`, `H083_coverage_le`, `H083_claim_holds`)
- **Claim**: Spectrum ratio max|F_hat|/sqrt(B) was measured generic (closed); the unrun angle is variance-to-mean of decompositions per target equalling 1.00+-0.03 for subspace and orbit bases, with any peak showing as overdispersion.
- **Prediction**: ell about $2^{28,B=2}12$: ratio 6.2+-0.5 for random and structured; variance/mean 1.00+-0.03.
- **Cheapest test**: Toy order about 2^28: BSGS logs, subspace/orbit/random FBs, sampled-t FFT, decomposition counts for 10^5 targets.
- **Falsifier**: Structured spectrum ratio >1.25, or variance/mean >1.1.
- **Lean gap (assumed, not proved)**: Variance/mean of decomposition counts equal to 1 and the flat Fourier spectrum of subspace/orbit bases on a real curve group are not proved; only the counting skeleton of m-fold sums on the exponent side is.

curve-selection (6)

H084 — Koblitz twin-curve amortisation: S3 independent of a

- **Handbook**: 16.4.1.3 Special curves, 16.4.2.2 Pairings and twists · **Prior art**: partially-tried · **Lean**: proved (`H084_claim`; theorems: `H084_partition`, `H084_amortised`, `H084_claim_holds`)
- **Claim**: One S3 solve over V gives relations for E_0 and E_1: about 50% pure E_0, 50% pure E_1, mixed <1%, so a paired workload amortises 2x; single-curve use only filters. Amortisation, not a single-instance gain.
- **Prediction**: n=23,a=1,b=1: split 50+-2%, mixed <1%; two-instance throughput 2.0x.
- **Cheapest test**: n=23,d=8: enumerate V^2, roots of S3, classify types by trace test, tabulate counts.
- **Falsifier**: Mixed >1%, split off 50/50 by >5%, or no two-instance workload exists in the protocol.
- **Lean gap (assumed, not proved)**: The no-mixed-solution hypothesis (mixed < 1%) and the 50/50 E0/E1 split from the trace test on Koblitz n=23 are curve facts assumed as hypotheses; the S3 solution set is an abstract list with a type tag.

H085 — Select GHS/Weil-descent covers by maximal rational-point count

- **Handbook**: 14.8.3 Niederreiter-Xing function fields · **Prior art**: partially-tried · **Lean**: proved (`H085_claim`; theorems: `H085_choose_step`, `H085_choose_mono`, `H085_choose_small`, `H085_surj_card`, `H085_claim_holds`)
- **Claim**: Relation probability per divisor class scales as N^g/(g!#J); top-decile-N genus-g covers yield >=3x the bottom decile, so choose covers by N_q(g) rather than at random.
- **Prediction**: 2000 genus-3 curves over F_8: Pearson r>0.95 with N^g/(g!#J); top/bottom decile 3-10x; q=16,g=4 upper bound about 69x.
- **Cheapest test**: Enumerate y^2+hy=f over F_8 (g=3); N and #J from L-polynomial; sumset of g-point sums via Cantor arithmetic; covered fraction vs formula.

- **Falsifier**: r<0.8, decile ratio <1.5, or the real GHS family offers no choice of N for the target.
- **Lean gap (assumed, not proved)**: The relation to the actual probability N^g/(g! #J) via L-polynomials and Cantor arithmetic, and surjectivity of the g-point sum map onto the Jacobian, are assumed; only binomial monotonicity and a surjection-cardinality bound are proved.

H086 — Invariant-dimension screen 3m>n+4 with rotation-only membership

- **Handbook**: 13.2.3 Matrix representations of finite fields, 13.2.4 Circulant matrices · **Prior art**: partially-tried · **Lean**: proved (`H086_claim`; theorems: `H086_orbit_closed`, `H086_countP_or`, `H086_countP_reps`, `H086_card_le`, `H086_claim_holds`)
- **Claim**: Primes n in 53..131 with ord_n(2)<=n/3 (73, 89, 113, 127) admit invariant subspaces of dimension m with 3m>n+4; orbit-distinct S3 counts there match generic subspaces within 10% while membership costs a few rotations/XORs instead of a matrix product.
- **Prediction**: n=23 (m=11,12) and n=31 (m=10,15): counts within +-10% of random V; membership 5-20x fewer word ops.
- **Cheapest test**: n=23,m=11: S3 over V^3 with x3 fixed on a few hundred values for invariant g vs 50 random V; time membership.
- **Falsifier**: Invariant count deviates >25% from random V, or membership gain <2x.
- **Lean gap (assumed, not proved)**: Existence of Frobenius-invariant subspaces of dimension m with 3m>n+4 for n=73,89,113,127, and that the orbit-distinct S3 count matches generic subspaces within 10%, are not proved; the orbit-size bound and orbit-key cover are hypotheses.

H087 — Volcano ascent to a j=0 or 1728 crater for an Aut-6 or Aut-4 quotient

- **Handbook**: 12.2.3 Isogenies, 12.2.8 Endomorphism ring and automorphism group, 12.2.9 Ordinary curves · **Prior art**: partially-tried · **Lean**: proved (`H087_claim`; theorems: `H087_orbit_six`, `H087_orbit_four`, `H087_transport`, `H087_claim_holds`, `H087_relation_ratio`)
- **Claim**: For smooth-conductor floor curves, walking to the crater (Aut order 6 or 4) lets the factor base close under zeta, cutting relations and LA size about 3x at equal base size; rho cost is level-independent, and random-curve targets are not covered.
- **Prediction**: p about $2^{30..2}40$, f=2 or 4: relations ratio (crater/floor) 0.33+-0.05; LA ratio about 0.11 dense; walk cost <1e-3.
- **Cheapest test**: 5 toy floor curves D=-12 or -48; Velu walk to j=0; same m=2 small-x IC with orbit quotient on both; random curve control.
- **Falsifier**: Relation ratio >0.5, transported log fails, or base cannot be made orbit-closed.
- **Lean gap (assumed, not proved)**: Existence of the Velu isogeny walk to a j=0/1728 crater, free action of Aut of order 6 or 4 on the factor base, and that the isogeny is a homomorphism of cyclic groups with image order dividing the source order are assumed (divisibility is a hypothesis of the transport lemma).

H088 — Invariant-subspace dimension menu from x^n-1 as a curve-size selector with yield test

- **Handbook**: 10.2.1 LFSR sequences, 10.2.3 Maximal period sequences, 15.1.3 Cyclic codes, 14.9.2 Weights of multiples of polynomials, 5.4.3 Cyclotomic modules · **Prior art**: tried-closed · **Lean**: proved (`H088_claim`; theorems: `H088_trivial_stable`, `H088_zero_rot_11_13`, `H088_dims_primitive`, `H088_ord_values`, `H088_claim_holds`)
- **Claim**: The census (2 primitive mod n gives only dims 0,1,n-1,n, including n=131) is closed. New angle: the explicit near-n/2 and near-n/3 dimension and count table for n=53..131, plus a measured yield and partial-stability (coordinate subspace column ratio >=0.97) test where such subspaces exist.
- **Prediction**: At most 25% of primes admit dim within 1 of n/3; n=127 has 7j and 7j+1 dims; coordinate subspaces at n=23 give columns/|V_I| in [0.97,1.0]; invariant yield equals generic within 10%.
- **Cheapest test**: Factor x^n-1 over F_2 for n=53..131; subset-sum divisor degrees; n=73 (d=24,25) invariant vs generic yield on a toy curve; n=23 coordinate-subspace orbit counts.
- **Falsifier**: A primitive-root prime with a stable subspace of dimension in [2,n-2], or columns/|V_I|<0.9 at n=23.
- **Lean gap (assumed, not proved)**: Semisimplicity of F_2[x]/(x^n-1) and the identification of stable subspaces with unions of cyclotomic cosets are taken as the hypothesis StableDim d <-> d in h088Dims sizes; the measured yield and column-ratio tests are not modelled.
- Merged from pool keys: P4-3, P14-0, P5-0

H089 — Berlekamp-Massey linear-complexity screen of Tr(x(kG)) for weak curves

- **Handbook**: 10.4.2 Linear complexity analysis, 10.4.4.1 Explicit sequences · **Prior art**: partially-tried · **Lean**: proved (`H089_claim`; theorems: `H089_lc_mono`, `H089_fold_congr`, `H089_determined`, `H089_bands`, `H089_claim_holds`)
- **Claim**: For smooth curves over GF(2^n) in ecdb, BM linear complexity of N=2048 terms of Tr(x(kG)) lies in N/2+-8; a nodal cubic gives L<=3n, validating the screen. An anomaly would be a lead.
- **Prediction**: At least 1000 smooth curves fall in [1016,1032]; nodal b=0 cubic over GF(2^21) gives L<=63; expected null.
- **Cheapest test**: For each ecdb curve with n<=40 and 5 generators: 2048 multiples, trace bit, BM over GF(2); nodal and supersingular controls; shuffled-sequence null.
- **Falsifier**: Positive control not detected (L>3n), or smooth curves outside the band above the 2^-16 rate.
- **Lean gap (assumed, not proved)**: Berlekamp-Massey itself is not implemented; only the soundness skeleton (an order-r recurrence is determined by r terms, complexity bound monotone) is proved. That Tr(x(kG)) on smooth curves has complexity N/2+-8 and nodal cubics have L<=3n is a curve fact not proved.

hardware (11)

H090 — ONB canonical orbit key with trace as parity (type-II n=53,131)

- **Handbook**: 16.7.4 Normal bases, 16.7.5 Optimal normal bases · **Prior art**: partially-tried · **Lean**: proved (`H090_claim`; theorems: `H090_canon_idem`, `H090_check_five`, `H090_order_131`, `H090_cost_gain`, `H090_claim_holds`)
- **Claim**: In a type-II ONB Frobenius is a bit rotation and Tr is parity, so orbit canonicalisation is the least rotation in about n bit-ops vs about n-1 squarings; only an end-to-end measured saving counts (repricing alone is accounting).
- **Prediction**: Canonicalisation 5-20% of collection cost in polynomial basis, <1% in ONB; end-to-end 1.05-1.25x at n=131; <5% share at n=53 means no gain.
- **Cheapest test**: Instrument the Frobenius-orbit IC at n=53,131: squarings and mults spent canonicalising as a fraction of per-relation cost at 0/0.1/1 GAE vs the ONB bit-op count.
- **Falsifier**: Canonicalisation <3% of per-relation cost at all prices, or ONB multiplication overhead gives end-to-end ratio <1.0 at 1 GAE.
- **Lean gap (assumed, not proved)**: That a type-II ONB at n=53,131 makes Frobenius a bit rotation and trace parity is assumed; rotation-invariance of the least-rotation canon is proved by decide only for n=5 and in general only as a hypothesis-based idempotence lemma, and the end-to-end measured saving (canonicalisation share of per-relation cost) is not modelled beyond Cost arithmetic.

H091 — GF2P8AFFINEQB bit-matrix applies for the linear maps of the lifted loop

- **Handbook**: 16.2.6 AES (Rijndael) · **Prior art**: untried · **Lean**: proved (`H091_claim`; theorems: `h091_xor_xor`, `H091_apply_linear`, `H091_comp_linear`, `H091_blocks_64`, `H091_blocks_131`, `H091_blocks_quadratic`, `H091_blocks_mono`, `H091_instr_mono`)
- **Claim**: On GFNI+AVX-512 hosts the linear-map part of the lifted loop reaches >=2x the candidates per cycle of 8-bit table lookups at n=64..131 (throughput- not load-bound); block count is quadratic in n/8.
- **Prediction**: Ratio 1.5-3x at n=64, below 1x at n>=131.
- **Cheapest test**: One n=64 and one n=131 F2-linear map over 2^20 vectors, table vs VGF2P8AFFINEQB, perf stat cycles on a GFNI host via isolab-eks; report the crossover n.
- **Falsifier**: Table lookup faster at n=64 or crossover below n=48; host without GFNI voids, not refutes.
- **Lean gap (assumed, not proved)**: Measured cycles per GFNI instruction vs per table lookup on a GFNI+AVX-512 host (throughput- not load-bound); H091_claim takes the 2x cycle ratio as a hypothesis.

H092 — Longest-zero-run canonicalisation of Frobenius orbits (bit-parallel)

- **Handbook**: 14.9.2 Cyclic structure F_2[X]/(X^n-1), necklaces · **Prior art**: tried-open · **Lean**: proved (`H092_claim`; theorems: `H092_le_lz`, `H092_min_has_max_run`, `H092_fast_eq_naive_upto6`, `H092_ops_131`, `H092_ops_ratio`)

- **Claim**: Minimal rotation starts at a maximal zero run; clz/ctz plus about 2 comparisons canonicalise a 131-bit word in O(n/64) ops, identical to naive. An earlier arm saved 8% instructions but did not resolve against the native limit.
- **Prediction**: At least 10x fewer cycles than naive; end-to-end >=1.2x if canonicalisation was >=20% of candidate cost; ties 1.3-2.2.
- **Cheapest test**: C microbenchmark on random 131-bit words, 10^8 calls; verify equality with naive on 10^7 words and all words for n<=24; count ties; perf-ipc where a PMU exists.
- **Falsifier**: Any output differing from naive, speedup <3x, or mean tied starts >4.
- **Lean gap (assumed, not proved)**: General-n proof that the fast run-start canonical form equals the naive minimum rotation is only exhaustive for n<=6 plus the exchange lemma (min rotation has max zero run); the word-op model is not measured cycles.

H093 — MDS-arc checksum protection for block Lanczos/Wiedemann on Spot

- **Handbook**: 14.4.6 k-Arcs and linear MDS codes · **Prior art**: untried · **Lean**: proved (`H093_claim`; theorems: `H093_cks_add`, `H093_cks_single`, `H093_detect_of_mds`, `H093_single_fault_detect`, `H093_overhead_le_2pct`, `H093_cheaper_than_duplicate`)
- **Claim**: With r=4 Vandermonde checksum rows c^(j)A, every corruption of <=4 coordinates per matvec is detected and <=2 corrected at <2% extra cost, vs about 100% for duplicate verification. Engineering saving.
- **Prediction**: Overhead 0.01-0.1% extra multiplies at dimension 2^16; detection 100% within bound; beats checkpoint-only by 1.2-2x at 5%/hour preemption.
- **Cheapest test**: Block Lanczos mod 31-bit l on a 2e4-dim sparse matrix, 20 nonzeros per row; inject 1-5 faults per 1000 matvecs; four arms: none, checkpoint, duplicate, MDS checksum.
- **Falsifier**: Any undetected corruption inside the bound, a miscorrection, or overhead >2%.
- **Lean gap (assumed, not proved)**: The Vandermonde/MDS property (any nonzero error pattern of weight 1..4 is seen by some checksum row mod l) and unit weights mod l are hypotheses; correction of <=2 faults is not modelled; overhead ratio depends on the measured check period q and row weight.

H094 — Latin-square XOR-skewed storage for conflict-free pair-sum table access

- **Handbook**: 14.1.1 Prime powers, 14.1.5 Affine and projective planes · **Prior art**: partially-tried · **Lean**: proved (`H094_claim`; theorems: `h094_xor_cancel`, `h094_xor_eq_zero`, `H094_row_conflict_free`, `H094_skew_injective`)
- **Claim**: Banking entry (i,j) at i XOR j over GF(2^5) gives conflict-free access along rows, columns and all slopes a!=1, removing shared-memory bank conflicts in the pair-sum kernel; odd-stride padding is already used elsewhere.
- **Prediction**: Replays drop from 2-4 per warp access to 0; kernel 1.1-1.3x if memory-bound.
- **Cheapest test**: CUDA kernel, B=2^12: row-major, padded, XOR-skewed layouts; ncu l1tex bank-conflict counters; cycles per pair.
- **Falsifier**: Speedup <5% or non-zero conflicts after skewing.
- **Lean gap (assumed, not proved)**: The GF(32) multiplication table is an abstract structure h094GF32 (distributivity, one, no zero divisors) rather than a concrete table; kernel bank-conflict counters and speedup are measured on GPU, not proved.

H095 — Projective-plane block design to shard all-pairs decomposition

- **Handbook**: 14.5.1 Block designs basics, 14.4.1 Projective spaces, 14.5.3 Singer difference sets · **Prior art**: partially-tried · **Lean**: proved (`H095_claim`; theorems: `H095_line_pairs`, `H095_work_sum`, `H095_total_work`, `H095_transfer_le`, `H095_p7_transfer`, `H095_claim_holds`)
- **Claim**: Placing shard pairs by lines of PG(2,p) (lambda=1) gives zero redundant pair work and per-worker transfer B(p+1)/M vs B for all-gather; total work identical.
- **Prediction**: p=7 (M=57): per-worker memory down about 7x; wall-clock 1.0-1.5x on Spot clusters.
- **Cheapest test**: Simulate 57 workers at B=2^14: bytes moved and work per worker for all-gather, random 2-D tiling and Singer design; check coverage equals C(M,2) exactly.
- **Falsifier**: Wall-clock gain <10%, or any shard pair covered zero or twice.
- **Lean gap (assumed, not proved)**: Existence of the PG(2,p) line blocks with lambda=1 (pair cover) is a hypothesis; the total-work and transfer arithmetic is proved from block count and block size; wall-clock gain on Spot clusters is empirical.

H096 — Reverse Cuthill-McKee reordering for block-Wiedemann matvec locality

- **Handbook**: 13.4.5.1 Reordering, 13.4.4.1 Wiedemann · **Prior art**: partially-tried · **Lean**: proved (`H096_claim`; theorems: `h096_finSum_ofFn`, `H096_finSum_perm`, `H096_relation_invariant`, `H096_cycles_ratio`)
- **Claim**: For B>=2^20 and 130-bit ell, RCM plus heavy-column grouping cuts L2 misses per nonzero >=30% and cycles per nonzero >=1.25x vs insertion order; a cycle-level, not op-count, gain.
- **Prediction**: 1.25-1.8x lower cycles per nonzero; <1.1x at B=2^16; IPC 0.6->0.9.
- **Cheapest test**: Synthetic B=$2^{14..2}$2, row weight 6, Zipf(1) column weights: insertion vs RCM vs degree sort, perf stat cycles and L2 misses on an isolab host, 20 paired repeats.
- **Falsifier**: Ratio <1.1 at B=2^22 with CI excluding 1.25, or no PMU and simulated miss reduction <10%.
- **Lean gap (assumed, not proved)**: The 30% L2-miss reduction from RCM ordering and the memory-bound fraction (2*compute <= miss stall) are measured hypotheses; only permutation invariance of relations and the cycle-model implication are proved.

H097 — Batch-affine vs projective candidate generation, binary vs prime split

- **Handbook**: 12.3.3 Coordinate systems, 12.3.4 Explicit formulas, 12.3.6 Char-2 short Weierstrass · **Prior art**: partially-tried · **Lean**: proved (`H097_claim`; theorems: `H097_mixed_le`, `H097_batchInv_le`, `H097_affine_le_proj`, `H097_ledger_ratio`)
- **Claim**: With k>=64 walks, batch-affine addition is >=1.6x faster per candidate than projective plus batch conversion in GF(2^131) with CLMUL, and <=1.1x on 255-bit Edwards. Prime side measured against batching; binary side unmeasured.
- **Prediction**: GF(2^131): 1.8-2.3x at k=64..256; GF(2^53): 1.3-1.6x; 2^255-19: 0.95-1.1x.
- **Cheapest test**: C microbenchmark, 1e7 additions per scheme with matched tables and seeds on isolab-eks; M/S/I ledger matching formulas; A/A arm; perf-ipc cycles.

- **Falsifier**: Binary speedup <1.15x at k=64, or prime-field speedup >1.3x.
- **Lean gap (assumed, not proved)**: Cycle cost is assumed proportional to the M/S/I ledger (CLMUL multiplication/squaring/inversion costs S<=M/10, I<=25M); the prime-field (<=1.1x) arm and real cycle ratio are unmeasured here.

H098 — Booth/Duval canonical key at n=131 against squaring chains

- **Handbook**: 12.2.8 Frobenius on Koblitz, 12.3.6 Char-2 arithmetic · **Prior art**: partially-tried · **Lean**: proved (`H098_claim`; theorems: `H098_orbit_count`, `H098_orbit_count_div`, `H098_key_rotation_invariant_upto6`, `H098_key_le_word_upto6`, `H098_orbits_n5`, `H098_orbits_n7`)
- **Claim**: Normal-basis keys are integrated (8.6x at degree 61); open is Booth/Duval at n=131: one basis change plus Booth is >=1.5x fewer cycles than min-over-squarings, with identical orbit partition.
- **Prediction**: n=131 AVX2: polynomial basis 1.3e3-2e3 cycles vs 4e2-8e2; speedup 1.6-3x; 1.0-1.3x at n=53.
- **Cheapest test**: C benchmark, 1e6 random x for n in {53,83,131}: min over squarings vs Booth on a normal-basis image; orbit ids via hash map; perf-ipc cycles; A/A.
- **Falsifier**: Speedup <1.15x at n=131, any orbit-partition mismatch, or basis change exceeds the saving.
- **Lean gap (assumed, not proved)**: H098_claim (key rotation invariance and key<=w for all words) is proved only exhaustively for n<=6; Booth/Duval equals naive minimum rotation, the free prime-order action at n=131, and the 1.5x cycle gain after basis change are not proved.

H099 — Itoh-Tsujii plus Montgomery batching as the affine-add primitive of the collection loop

- **Handbook**: 11.1.8.2 Inversion in extension fields, 11.1.5.2 Multiplication in extension fields · **Prior art**: partially-tried · **Lean**: proved (`H099_claim`; theorems: `H099_itoh_131`, `H099_itoh_83`, `H099_itoh_53`, `H099_batchInv_20`, `H099_breakeven_131`, `H099_batch_beats_ld_131`, `H099_claim_131`)
- **Claim**: Batched affine addition (about 5 mults) with Itoh-Tsujii (8 mults at n=131) beats Lopez-Dahab mixed addition (8-11 mults) per point-sum at batch >=16 for n in 53..131, no Z coordinate.
- **Prediction**: 1.3-1.8x at batch 16-64 at n=131; break-even batch 4-8 when squaring is about 0.1 mult.
- **Cheapest test**: GF(2^n), n in {53,83,131}: affine+batch+Itoh-Tsujii vs LD mixed add, interleaved, A/A, batch 1,4,16,64; ns and mults per add.
- **Falsifier**: LD mixed add at least as fast at batch 16 for n=131, or break-even batch >64.
- **Lean gap (assumed, not proved)**: The Itoh-Tsujii multiplication count formula (floor(log2(n-1)) + wt(n-1) - 1) and the LD mixed-add cost 8M+5S are assumed; H099_claim for general n in 53..131 is stated, only n=131 is proved; real ns per add is empirical.

H100 — Rotation-based membership test g(sigma)x=0 with sparse generator

- **Handbook**: 5.3.5 Fast arithmetic using normal bases, 5.2.1 Normal bases · **Prior art**: partially-tried · **Lean**: proved (`H100_claim`; theorems: `H100_rotPow_eq`, `H100_rot_pow_n`, `H100_apply_rot`, `H100_apply_xor`,

`H100_kernel_xor_closed`, `H100_kernel_rot_invariant`, `H100_cost_ratio`, `H100_claim_holds`, `H100_divides_x7`, `H100_divides_x15`)

- **Claim**: Membership in V=ker g(sigma) costs wt(g) XORs of rotations (normal basis), 8-15x cheaper than the polynomial-basis matrix test, but whole-loop gain is only 1-5% since roots and inversions dominate; likely null.
- **Prediction**: Membership 8-15x fewer cycles; loop 1-5%; net <1.05x or negative if multiplication must run in the normal basis.
- **Cheapest test**: n=127, V dim 63 or 7*9: membership microbenchmark in both bases (1e8 candidates), then the n=31 toy S3 pair loop; perf-ipc where available.
- **Falsifier**: Full-loop speedup >1.2x (then it matters), or membership speedup <3x.
- **Lean gap (assumed, not proved)**: Normal-basis representation of Frobenius as rotation, g | x^n-1 for the chosen n=127 generator (checked by decide only for n=7 and n=15), and the whole-loop speedup (1-5%, likely null) are not proved.